

Implementation and Verification of a CAN/CAN FD Protocol Controller in VHDL

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Abstract

This thesis describes the design, implementation, and verification of a CAN/CAN FD protocol controller in VHDL-93 for Everllence's FPGA-based IO-extender platform. The controller complies with ISO 11898-1 and supports CB, CE, FB, and FE frame formats with dual bit rate switching and Transmitter Delay Compensation for the FD data phase. The design follows the ISO 11898-1 layered reference model, with the MAC, PCS, and FCE sub-layers implemented as independently testable modules and connected to the host via Avalon-ST interfaces.

37 requirements were derived from ISO 11898-1, each linked to its source clause, verification method, and testbench assertion. 28 are closed against passing testbenches or code inspection. The nine open requirements comprise six LLC-layer requirements deferred pending LLC sub-layer implementation, one not applicable to this architecture, one optional operational feature, and one requiring a frame-aware reference model for error-type-specific injection coverage.

The design was synthesized on a Cyclone 10 LP FPGA target using 4,608 logic elements at 30% device utilization - 4.0× the existing CAN Classic controller. The worst-case f_{max} of approximately 127 MHz exceeds the highest recommended CAN FD system clock of 80 MHz by more than a factor of 1.5.

Approval

This thesis was prepared at the Department of Applied Mathematics and Computer Science (DTU Compute), Technical University of Denmark, in partial fulfilment of the requirements for the degree of Bachelor of Science in Engineering. The work was carried out in collaboration with Everllence, where the controller is intended for use in high-reliability engine controller applications.

It is assumed that the reader has a working knowledge of digital logic design and binary communication protocols. Familiarity with VHDL or a similar hardware description language is an advantage but is not strictly required to follow the architectural and verification discussions.

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Abbreviations

Table 1: Abbreviations used in this report.

Abbreviation	Meaning
ACK	Acknowledgment
AD	ACK Delimiter
AEF	Active Error Flag
AI	Artificial Intelligence
BRS	Bit Rate Switch
CAN	Controller Area Network
CANH	CAN High bus wire
CANL	CAN Low bus wire
CB	Classic Base (frame format)
CBFF	Classic Base Frame Format
CC	CAN Classic
CD	CRC Delimiter
CE	Classic Extended (frame format)
CEFF	Classic Extended Frame Format
CRC	Cyclic Redundancy Check
D	Dominant
DF	Data Frame
DLC	Data Length Code
DMA	Direct Memory Access
DUT	Device Under Test
ED	Error Delimiter
EOF	End of Frame
ESI	Error State Indicator
FB	FD Base (frame format)
FBFF	FD Base Frame Format
FCE	Fault Confinement Entity
FD	Flexible Data Rate
FDF	FD Frame bit
FE	FD Extended (frame format)
FEFF	FD Extended Frame Format
FPGA	Field-Programmable Gate Array
FSB	Fixed Stuff Bit
FSM	Finite State Machine
FTYP	Frame Type
HDL	Hardware Description Language
IDB	Identifier Base field (11-bit base ID)
IDE	Identifier Extension bit
IEXT	Identifier Extension field (18-bit extended ID)
IFS	Inter Frame Space
INT	Intermission

Abbreviation	Meaning
IP	Intellectual Property
IPT	Information Processing Time
ISO	International Organization for Standardization
LE	Logic Element
LLC	Logical Link Control
LLM	Large Language Model
MAC	Medium Access Control
MIT	Massachusetts Institute of Technology (license)
MSB	Most Significant Bit
OD	Overload Delimiter
OF	Overload Flag
OSVVM	Open Source VHDL Verification Methodology
PCS	Physical Coding Sublayer
PE	Phase Error
PEF	Passive Error Flag
PS	Propagation Segment (PROP_SEG)
PS1	Phase Segment 1 (PHASE_SEG1)
PS2	Phase Segment 2 (PHASE_SEG2)
R	Recessive
RF	Remote Frame
RRS	Reserved Remote Request Substitution bit (FD frames)
RTL	Register Transfer Level
RTR	Remote Transmission Request
RX	Receiver / Receive
SB	Stuff Bit
SBC	Stuff Bit Count
SJW	Synchronization Jump Width
SOF	Start of Frame
SP	Sample Point
SRR	Substitute Remote Request
SS	Sync Segment (SYNC_SEG)
SSP	Secondary Sample Point
ST	Suspend Transmission
TDC	Transmitter Delay Compensation
TEC/REC	Transmit Error Counter / Receive Error Counter
TQ	Time Quantum
TX	Transmitter / Transmit

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1 Introduction

Everllence (formerly MAN Energy Solutions) is a provider of propulsion and decarbonization solutions for the marine and energy industries, designing large two-stroke and four-stroke combustion engines for ship propulsion and power generation. This thesis was written within the engine controller division, where FPGA-based control hardware is developed and maintained for integration into Everllence's engine platforms. Industrial control systems for large marine engines demand communication protocols that combine fault tolerance and confinement, multi-master arbitration, and multi-decade service reliability. The Controller Area Network (CAN) communication protocol, developed by Bosch in the 1980s for automotive and industrial control, meet these basic demands [1].

The original version of CAN protocol, CAN Classic (CC), forms the basis of Everllence's existing CAN infrastructure, which is implemented through a VHDL CC protocol controller deployed on an IO-extender board forming part of the Triton motor controller system. As control systems grow more data-intensive, the eight-byte payload and 1 Mbit/s ceiling of CC has become a practical constraint. The CAN FD (Flexible Data Rate) protocol extension, introduced in 2012 [2], extends the maximum data payload to 64 bytes and raises the bit rate beyond 8 Mbit/s while preserving the arbitration and fault-confinement/tolerance architecture of CC. Thus, CAN FD (CF) is the natural migration path for Everllence's existing CAN infrastructure.

1.1 Background

This section establishes the technical context for the thesis work. CC and CF are both governed by ISO 11898-1 [3], which defines the CAN data link layer protocol. The subsections cover each at the level of motivation and key properties, and introduce the existing Everllence CC controller (`can_bus_controller`) that provided the immediate starting point for the redesign.

1.1.1 CAN Classic

CC implements a serial multi-master bus connecting electronic control units in automotive environments. CAN uses a shared two-wire differential bus on which all nodes broadcast simultaneously and arbitrate access without any designated bus master. Any node may initiate a transmission at any time. Contention is resolved by a non-destructive bitwise arbitration in which the transmitter with the lower-priority identifier detects the collision and silently withdraws, leaving the winner's frame intact. The bus uses two conductors, CANH and CANL, whose voltage difference encodes the bit value, see Figure 1. Twisting the conductors ensures that external electromagnetic interference couples equally onto both wires. Because the receiver measures only the differential voltage, this common-mode noise cancels - a practical necessity in the electrically harsh environment.

CC's error-handling architecture is a distinguishing feature relative to simpler serial protocols. Five complementary error detection mechanisms operate concurrently on every transmitted frame: transmitter bit monitoring, frame format checking, cyclic redundancy checking, acknowledgment checking, and bit stuffing violation detection. The residual error probability, the probability that a corrupted frame passes all detection mechanisms undetected, for an eight-byte frame in a ten-node network is on the order of 10^{-9} per frame - several orders of magnitude lower than contemporary automotive bus alternatives such as VAN and SCP [4]. A fault confinement mechanism tracks each node's error history and automatically disconnects persistently faulty nodes from the bus without disrupting communication between healthy nodes. Together these properties made CC the protocol of choice for safety-relevant in-vehicle networks.

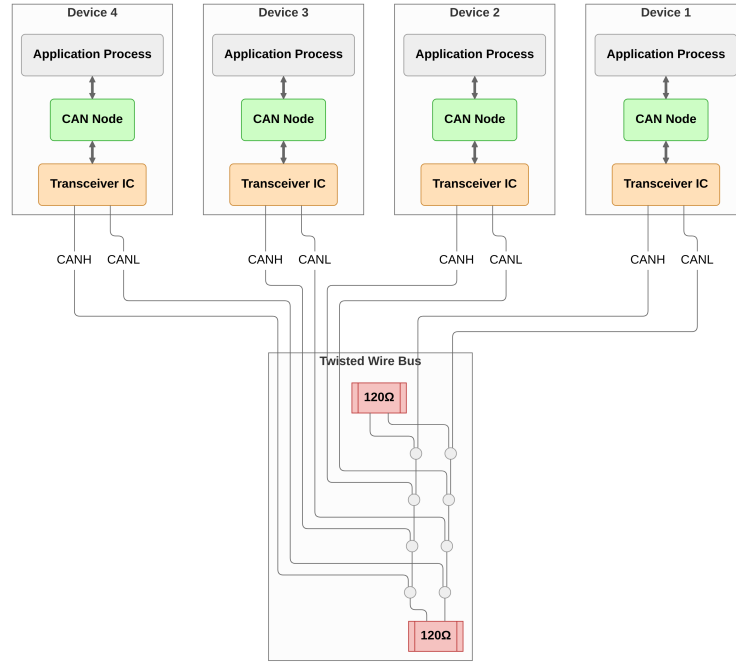


Figure 1: Four CAN nodes connected to a shared twisted-wire bus via CANH and CANL conductors. Each node comprises an application process, a CAN controller, and a transceiver IC. The bus is terminated with 120Ω resistors at each end.

1.1.2 CAN FD

The design goal of CF was to extend payload and bit rate while preserving the arbitration and fault-confinement architecture that distinguished CC from its contemporaries. CC's original data payload was capped at eight bytes per frame, limiting raw throughput to around 1 Mbit/s. As embedded control applications became more data-intensive, this ceiling became a practical constraint. CF extends the maximum payload to 64 bytes. In addition, it introduces a dual bit rate frame format comprising two phases: a higher-speed data phase with bit rates of 8 Mbit/s or beyond, and a nominal arbitration phase operating at CC bit-rates.

At high data phase bit rates the transceiver's TX-to-RX loop delay (approximately 100 ns for a typical CF transceiver such as the TCAN1042 [5]) may exceed the data phase bit time. Since the fault confinement mechanism relies on transmitter nodes monitoring their own transmitted bits, this necessitates the Transmitter Delay Compensation (TDC) mechanism introduced in CF, see Section 3.3. For data-phase to arbitration-phase bit rate ratios below approximately 9, CF accepts the same oscillator tolerance as CC [6]. Below this threshold, the arbitration-phase resynchronization condition remains the binding constraint, so the higher data-phase bit rate does not impose a tighter oscillator requirement.

CF also strengthens the error detection architecture relative to CC. CC's CRC-15 polynomial preserves a Hamming distance of 6 only up to a certain frame length, making it insufficient for CF's longer payloads. CF replaces it with a 17-bit polynomial for frames up to 16 data bytes and a 21-bit polynomial for frames up to 64 bytes, preserving the Hamming distance of 6 across the extended payload range and guaranteeing detection of any five or fewer bit errors [2].

CF also addresses a gap in CC's stuff-bit handling: in CC, dynamic stuff bits are excluded from the CRC calculation, allowing rare two-bit errors that create or destroy a stuff condition to go undetected. CF closes this by including dynamic stuff bits in the CRC data feed and introducing the Stuff Bit Count (SBC) field as

an independent check on the number of inserted stuff bits. The combined effect of these changes extends the CRC field from 16 bits in CC to 28 bits (CRC17) or 33 bits (CRC21). With 12 or 17 more bits required to accidentally match the expected CRC field, the residual error probability for this class of faults is reduced by several orders of magnitude [7].

1.1.3 Everllence's Current CAN Controller

The starting point for this project is Everllence's internally developed current CC controller (`can_bus_controller`). The controller is implemented in VHDL and has been integrated into a production IO-extender FPGA design. It supports CC frames with both 11-bit (base) and 29-bit (extended) identifiers at bit rates up to 500 kbit/s, and has been verified through hardware bring-up on physical CAN buses. The initial version was developed by the author of the present document during an internship at Everllence and has since been extensively modified by other engineers at the company.

The top-level wrapper for `can_bus_controller` instantiates a combined TX/RX frame FSM, a bit timing generator, a dynamic bit stuffer, a CRC-15 engine, and two Avalon-ST converters for frame serialization and deserialization.

The central component is the orchestrating FSM that handles both transmission and reception in a single process. It manages frame arbitration, bit-level TX and RX, stuff-bit error checking, CRC validation, ACK handling, and error flag generation. Fault confinement (error active, error passive, bus off node state transition) is handled in a separate process within the main FSM entity.

While the `can_bus_controller` is functional for CC, several areas of its design would require significant rework to support CF. The main limitations are listed below.

- **Single bit rate domain:** The controller operates at a single bit rate with no data-phase switching or TDC support (Section 3.3).
- **Dynamic bit stuffing only:** The stuffer has no mechanism for fixed bit stuffing or SBC generation (Section 3.4).
- **Single CRC engine:** The controller has a single CRC engine, where CF compliance requires three running in parallel with separate data feed paths (Section 3.5).
- **Coupled fault confinement:** Fault confinement is coupled into the frame FSM with no clean sub-layer boundary (Section 3.6). ISO 11898-1 defines it as a distinct cross-cutting entity, and separating it improves both conformance to the standard's reference model (Section 3.1) and testability.
- **Format-dependent frame structure:** CC and CF frames diverge in the control and data phases, where CF introduces format-specific fields with no Classic equivalent (Section 3.2). The existing FSM has no clean mechanism to handle this divergence across four frame variants.

The changes required across bit timing, bit stuffing, CRC, and frame format handling are pervasive enough to justify a clean-slate redesign.

1.2 Available Third-Party CAN FD IP Cores

Before committing to an in-house redesign, the available CF controller IP cores were evaluated. Table 2 summarizes the candidates, spanning both open-source and commercial offerings.

1.2.1 Open-Source Implementations

CTU CAN FD [8] is the only mature open-source CF controller available as synthesizable HDL. Developed at the Czech Technical University in Prague, it is written in VHDL, licensed under MIT, and has been conformance-tested against ISO 16845-1 [9]. The controller includes a full TX and RX pipeline with up to four TX buffers, acceptance filtering, timestamping, and a register interface with DMA support.

1.2.2 Commercial Implementations

Bosch M_CAN [10] is the reference CF controller developed by Bosh. M_CAN is the IP core embedded in most automotive micro controllers. It is licensed under a non-disclosure agreement with per-design royalty fees.

AMD/Xilinx CAN FD [11] is a soft IP core included in the Vivado Design Suite. It provides an AXI4-Lite register interface with up to 32 acceptance filters, TX mailboxes, and RX FIFOs. It is device-locked to AMD/Xilinx FPGAs and cannot be ported to other targets.

Technology-independent RTL cores. CAST CAN FD [12] is a technology-independent CF IP core licensed per-design with an upfront fee. Major EDA vendors including Synopsys (DesignWare) and Cadence offer similar ASIC-targeted cores under commercial licensing programs.

Table 2: Survey of available CF controller IP cores.

Implementation	Source	License	Scope	Conformance Tested
CTU CAN FD [8]	VHDL (MIT)	MIT	Full node	Yes
Bosch M_CAN [10]	Closed	Per-design royalty	Full node	Yes (reference)
AMD CAN FD [11]	Closed	Vivado-included	Full node	Yes
CAST CAN FD [12]	Closed	Per-design fee	Full node	Yes

1.2.3 Rationale for In-House Development

None of these solutions satisfies Everllence’s combined requirements. The disqualifying factors span IP ownership, verification authority, architectural scope, integration with existing infrastructure, and platform independence - each addressed in turn below.

- **IP ownership and supply chain independence:** Everllence’s engine controllers carry service commitments of up to thirty years. Commercial IP cores introduce a licensing dependency on an external vendor over that full horizon - vendors may discontinue support, change licensing terms, or be acquired. Owning the RTL outright eliminates this exposure and ensures that the design can be maintained, ported, and modified without third-party approval for the full product lifetime. The open-source CTU CAN FD avoids the licensing risk, but using it still means adopting a codebase whose architecture, naming conventions, and design decisions were made for a different context.
- **Verification authority:** In safety-critical domains, the verification evidence must be traceable from standard requirements to RTL assertions and testbench results. Adopting a third-party core means inheriting its verification artifacts rather than producing them. Everllence’s verification methodology requires full control over the verification plan, the testbench architecture, and the assertion coverage.
- **Architectural scope:** All available IP cores implement a complete CAN node - TX and RX pipelines, message memory, acceptance filtering, buffer management, register interfaces, and in

some cases DMA controllers. Everllence’s application requires only the protocol engine - the module that converts between byte-level frame data and the serial bus - which is also the scope of `can_bus_controller`. The higher-level buffering and filtering logic already exists in Everllence’s FPGA infrastructure. Adopting a full-node IP core would introduce unnecessary complexity and area overhead, and stripping the unused subsystems to fit the existing architecture offsets the benefit of using a pre-built core.

- **Integration with existing infrastructure:** Everllence’s FPGA designs use a specific Avalon-ST streaming interface for inter-module communication and established conventions for signal naming and module boundaries. A third-party core would require an adaptation layer to bridge its native interface to the existing infrastructure. An in-house design would use Everllence’s interface conventions natively, eliminating this integration overhead.
- **Platform independence:** The AMD/Xilinx CAN FD core is locked to Xilinx devices. The Bosch M_CAN and other commercial cores are delivered as technology-specific netlists or encrypted HDL for a particular target. The in-house design is written in portable VHDL-93, synthesizable on any FPGA platform ensuring that the IP remains usable if Everllence changes FPGA vendors.

1.3 Problem Statement

The architectural limitations of the existing controller (Section 1.1.3) and the unsuitability of available third-party IP cores (Section 1.2.3) together motivate a clean-slate CF protocol controller conforming to ISO 11898-1. No existing solution combines full IP ownership, a targeted data-link-layer scope matching Everllence’s integration requirements, and native compatibility with Everllence’s Avalon-ST interface conventions and VHDL Code Standard. The design described in this report addresses that gap directly.

1.4 Objectives

1. Implement a CC/CF protocol controller in VHDL, compliant with ISO 11898-1 [3].
2. Establish a structured requirements framework with traceability from ISO 11898-1 to testbench evidence, covering all implemented modules.
3. Produce an RTL design integrated via Avalon-ST interfaces into Everllence’s existing FPGA infrastructure.

2 Requirements

This section establishes the constraints that bound the design before any architectural decisions are made. Everllence’s coding standard, tooling, and existing FPGA infrastructure set the implementation framework, while the protocol requirements derived from ISO 11898-1 define the functional obligations of the controller.

2.1 Code Standard, Tooling, and Design Constraints

2.1.1 Tools and Language

The RTL source is implemented in VHDL-93. Everllence’s synthesis toolchain uses Quartus Prime [13], which does not fully support VHDL-2008 constructs in synthesis, making VHDL-93 the practical upper bound for synthesizable RTL. Testbenches are written in VHDL-2008 for the OSVVM verification framework [14]. SystemVerilog with UVM is the dominant industry alternative for RTL implementation and verification at this scale. The choice here follows company convention rather than a project-level technical

comparison. Riviera-PRO [15] is used for simulation. Sigasi [16] is used for linting and language-aware editing. Waveform figures are captured in GTKWave [17], timing diagrams are drawn in WaveDrom [18], and architecture diagrams in Mermaid [19].

Claude Code [20] was used for prose editing and VHDL review. Technical content, design decisions, and all source files are the author's own work.

2.1.2 Integration Requirements

1. **Avalon-ST user interface:** The CAN controller's external interface to the host system must use the Avalon-ST streaming protocol [21] (data, valid, ready, sop, eop). The requirement applies specifically to the boundary between the CAN controller and its user.
2. **IP library CRC block:** Everllence maintains a reusable, parameterised CRC generator (`gen_crc`) in its IP library. This module must be used for all CRC computations.

2.1.3 VHDL Code Standard

1. **Entity port types:** Entity ports are restricted to `std_logic`, `std_logic_vector`, and records or arrays of these types. Modes are restricted to `in` and `out`.
2. **Naming conventions:** A mandatory prefix/suffix scheme applies to all VHDL identifiers: types (`t_`), constants (`c_`), generics (`gc_`), processes (`p_`), functions (`f_`), packages (`pk_`), state variables (`s_`), entity inputs (`_i`), and entity outputs (`_o`).
3. **RTL design rules:** Synchronous processes must be sensitive to the clock only. Reset must be synchronous and initialize all control registers. FSMs are preferably implemented as single-process designs where all signal assignments are derived from the current state.
4. **Testbench requirements:** Testbenches must follow a black-box testing model and test cases must be ordered as: reset tests first, then a normal-usage test, then all remaining tests.

2.2 From Specification to Structured Requirements

The requirements engineering process addressed two key objectives [22]:

1. Extracting a clear and actionable set of requirements that could serve as a starting point for the design phase.
2. Establishing a clear, traceable link between the ISO 11898-1 specification and the verification environment.

Both objectives are complicated by the source material: normative requirements are distributed across subsections, bundled into compound clauses, and repeated from transmitter and receiver perspectives, interspersed with informative rationale prose.

The AI-augmented pipeline shown in Figure 2 was designed to address these extraction challenges systematically. The first step was converting the ISO 11898-1 PDF to Markdown - a format that can be efficiently searched and ingested by LLMs. The resulting Markdown file was then fed to a Claude Sonnet 4.6 LLM agent, which was prompted to extract all normative statements - sentences containing words like "shall", "should", "must", and their corresponding negations.

This process yielded a raw set of 168 normative statements linked to the ISO standard sections from which they were extracted. The normative set was then manually reviewed, consolidated, and distilled into a final set of 37 requirements (reproduced in Section C).

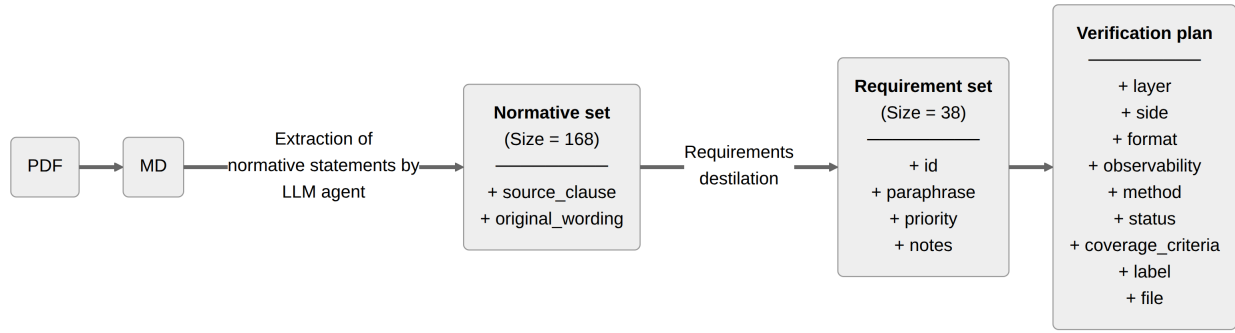


Figure 2: Pipeline generating the `verification_plan.toml` artifact from the ISO 11898-1 standard.

The requirements set is stored as a TOML file, one entry per requirement. Direct LLM editing of large structured files is unreliable - prone to silent entry deletion, field hallucination, and syntax corruption. To make iterative AI-assisted refinement of the plan viable, a custom Model Context Protocol (MCP) server was developed alongside it. The server exposes query, insert, update, and delete operations, together with bulk update and statistics utilities, as structured tool calls. Each write operation targets a single requirement entry and validates field values against the schema before committing. This bounds any model error to one requirement and prevents malformed data from reaching the file. Each requirement entry contains the following fields:

- **source_clause:** Links every requirement back to the ISO 11898-1 clause from which it was distilled, enabling the requirements set to be audited against the standard.
- **original_wording:** Verbatim ISO text for the relevant clauses. Preserving the source wording prevents paraphrase drift and provides a fallback for resolving ambiguity during implementation.
- **paraphrase:** A concise, implementer-facing restatement of the requirement. Where the ISO prose bundles multiple obligations into a single clause, the paraphrase enumerates them as numbered sub-claims, each independently verifiable.
- **priority:** A three-level rating - P1 (need-to-have, derived from “shall” obligations and core correctness), P2 (judged non-critical for this application: the bus functions in Everllence’s target use case without these, though they are part of full standard compliance), or P3 (optional, derived from “should” clauses or implementation-dependent features). The priority drove the implementation sequence and scope decisions.
- **notes:** Any residual clarifications not captured by the paraphrase - implementation constraints, out-of-scope markers, or known ambiguities. May be empty.

Of the 37 requirements, 30 are rated P1, four are P2, and three are P3. Each non-P1 rating reflects a deliberate scoping decision - the rationale for each is given in Table 3.

Table 3: Priority demotion rationale for all requirements not rated P1.

ID	Topic	Priority	Demotion rationale
REQ-002	LLC TX request and abort timing	P2	The 2-SOF processing window is a responsiveness guarantee, not a correctness constraint. A node that transmits eventually but outside this window sends valid frames.
REQ-011	Remote frame	P2	A data-only node is a valid CC/CF implementation. Remote frame support is a distinct feature subset not required for basic interoperability.

ID	Topic	Priority	Demotion rationale
REQ-015	ESI bit transmission	P2	ESI communicates the node's error state as an informational signal. Incorrect ESI does not abort a frame or trigger a protocol error at any receiver.
REQ-035	Error signaling enable	P2	Error signaling itself is covered by P1 requirements. This requirement concerns only the existence of a configurable disable mode, which is an optional operational feature.
REQ-004	Frame acceptance filtering	P3	Acceptance filtering is absent from Everllence's current controller, so no regression concerns. The only protocol-relevant filter - suppressing loopback of transmitted frames - is covered by the design.
REQ-034	Shared memory consistency	P3	The "may" language makes shared memory use optional.
REQ-036	DLC padding	P3	Padding with 0xCC applies only when the implementation exposes a configurable maximum-data-byte restriction. The feature may be waived entirely if that restriction is not implemented.

The distillation step that followed - consolidating 168 raw normative statements into 37 prioritized, independently verifiable requirements - required substantial manual effort that the AI could not replace. Deciding which statements address the same underlying obligation, how to bound each requirement so that it is independently verifiable, and how to assign priority in a way that is defensible against the standard text are judgment calls that depend on understanding the protocol at an implementation level. REQ-026 illustrates the challenge well - the corresponding extracted normative statements are:

1. *"Only one synchronization within one bit time (between two sample points) shall be allowed. After an edge is detected, synchronizations shall be disabled until the next time the bus state detected at the sample point is recessive."*
2. *"An edge shall cause synchronization only if the bus state detected at the previous sample point was recessive. If a transmitter uses transmitter delay compensation, also the first detected edge from recessive to dominant after the sample point of the CRC delimiter shall cause synchronization. An edge with a positive phase error shall not cause synchronization in a node sending a dominant bit."*
3. *"Hard synchronization shall be performed: on edges during inter-frame space (with the exception of the first bit of intermission); when a node is in bus-integration state; at the edge between the FDF bit and the following dominant res bit inside an FD frame; when a node is a receiver of an XL frame, at the edge between the XLF bit and the following dominant resXL bit; at the edge between the DH1 and DH2 bits and the DL1 bit; at the edge between the DAH or AH1 bit and the AL1 bit."*
4. *"All other recessive-to-dominant edges fulfilling rules a) and b) shall be used for resynchronization with one exception: a node transmitting an FD frame or an XL frame shall not synchronize while it transmits the data phase of that frame."*
5. *"After a hard synchronization, the bit time shall be restarted with Sync_Seg completed. Hard synchronization shall not be limited by the synchronization jump width."*
6. *"When the magnitude of the phase error is less than or equal to the synchronization jump width, the effect of a resynchronization shall be the same as a hard synchronization."*
7. *"When the magnitude of the phase error is larger than the synchronization jump width: if positive, Phase_Seg1 shall be lengthened by the synchronization jump width; if negative, Phase_Seg2 shall be shortened by the synchronization jump width."*

Statements 3 and 4 each embed CAN XL elements within otherwise in-scope obligations, which must be excluded without dropping the surrounding CB, CE, FB, and FE rules. The remaining statements interleave hard sync and resync rules across four sub-clauses, requiring the two mechanisms to be separated and their interaction made explicit. The resulting paraphrase is presented below, with the remaining requirement fields in Table 4.

Paraphrase:

1. Phase error: the time interval between an R-to-D edge and the Sync_Seg boundary of the current bit time. Positive when the edge falls after Sync_Seg (late), negative when before (early).
2. Qualifying edge: R-to-D, previous SP is R.
3. SJW is the maximum per-resynchronization adjustment to Phase_Seg1 or Phase_Seg2. Qualifying edges cause resynchronization. If $|\text{phase error}| \leq \text{SJW}$: same effect as hard synchronization. If $\text{phase error} > \text{SJW}$: $\text{Phase_Seg1} += \text{SJW}$. If $\text{phase error} < -\text{SJW}$: $\text{Phase_Seg2} -= \text{SJW}$.
4. One synchronization per bit time, re-enabled after the next R SP.
5. Hard synchronization: IFS edges (except first intermission bit), bus-integration state, and FDF-to-res transition in FD frames. Restarts bit time with Sync_Seg completed.

Table 4: REQ-026 distilled from seven extracted normative statements.

Field	Value
ID	REQ-026
Source	§7.3.5.1, §7.3.5.2, §7.3.5.3, §7.3.5.4, Figure 33
Priority	P1
Notes	RTL is stricter than ISO: sync is suppressed unconditionally when transmitting. Safe since the transmitter is the timing source.

3 CAN Classic and CAN FD Protocol Overview

The 37 requirements distilled in Section 2 define what must be implemented and verified - but they also function as a structured map to the protocol, since every requirement points to a mechanism that must be understood before implementation can begin. Those mechanisms - the sub-layer model, frame formats, bit timing, bit stuffing, CRC, and error handling - are covered here, each cross-referenced to the relevant REQ-NNN entries. Readers familiar with ISO 11898-1 may skip to Section 4.

3.1 Layered Reference Model

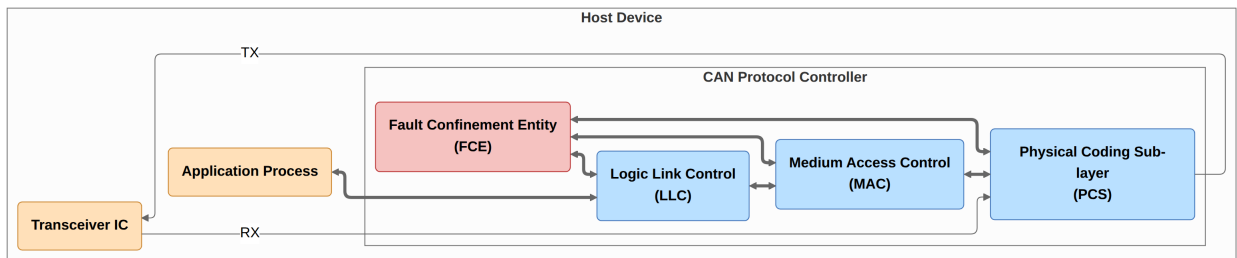


Figure 3: ISO 11898-1 CAN node reference model showing the LLC, MAC, PCS sub-layers and cross-cutting FCE.

ISO 11898-1 structures the CAN node reference model into three functional sub-layers - LLC and MAC in the data link layer, PCS in the physical layer - and a cross-cutting Fault Confinement Entity (FCE) [3, Fig. 4] (see Figure 3):

- **LLC (Logical Link Control):** Acceptance filtering (selecting received frames by identifier), overload notification (delaying the next frame when internal conditions require it), and recovery management (retransmission on error or lost arbitration), and supplying frames to the MAC.
- **MAC (Medium Access Control):** Encodes and decodes the frame bit-by-bit, performing bit stuffing and destuffing, CRC generation and checking, error detection and signalling, acknowledgment handling, and medium access arbitration.
- **PCS (Physical Coding Sublayer):** Bit timing and bus sampling (segmenting each bit time and reading the bus at the sample point), clock synchronization, and the TX/RX interface to the physical transceiver.
- **FCE (Fault Confinement Entity):** Escalating the node's error state from error active through error passive to bus off as transmit and receive error counts accumulate.

3.2 Frame Types and Formats

CAN frames may carry either an 11-bit base identifier or a 29-bit extended identifier, giving four frame formats: CBFF (CAN Classic Base Frame Format), CEFF (CAN Classic Extended Frame Format), FBFF (CAN FD Base Frame Format), and FEFF (CAN FD Extended Frame Format) (REQ-037). The four frame formats are depicted in Figure 4 along with the Active/Passive Error Flags (AEF/PEF). CBFF and CEFF additionally support Remote Frame (RF) variants, giving six frame format types in total.

All frame formats opens with a D SOF bit that triggers hard synchronization (Section 3.3) in all receiving nodes, followed by the arbitration, control, data, and CRC fields, an ACK slot, and a seven-bit EOF delimiter. The RTR bit is D for data frames and R for RF. The IDE bit distinguishes base frames from extended frames. Fixed-polarity form bits (SRR, r0, r1) carry no protocol instruction. The DLC encodes the number of data bytes in the payload (REQ-031). The ACK slot carries a dominant bit driven by every receiver that has validated the CRC. Each frame is followed by the interframe space - intermission (INT), suspend transmission (ST, error passive transmitters only), and bus idle (REQ-008).

CFFF share the same arbitration phase structure, with the FDF bit signalling an CF format when R. The CF control field contains a reserved form bit (res) alongside BRS and ESI. The DLC retains its 4-bit width but uses a non-linear mapping above 8 bytes, extending the maximum payload to 64 bytes (REQ-031). The BRS (Bit Rate Switch) bit controls the transition to the data-phase bit rate. When BRS is R the bus switches to the faster data rate immediately after the BRS sample point and returns to the nominal rate at the CRC delimiter (REQ-030). The ESI (Error State Indicator) bit reflects the transmitting node's fault-confinement state: a node in error passive state shall transmit ESI recessive (REQ-015). The CFFF CRC field is additionally prefixed by a Stuff Bit Count (SBC) - a Gray-coded count of dynamic stuff bits with a parity bit (REQ-016).

3.3 Bit Timing

Every CAN bit period is divided into four non-overlapping time segments measured in Time Quanta (TQ), where one TQ equals the system clock period multiplied by a programmable integer prescaler, see Figure 5. CF extends CC with an independently configured data-phase bit rate. A CF node therefore maintains two independent sets of segment parameters, one for the nominal rate and one for the data rate (REQ-024). The segments are listed described below.

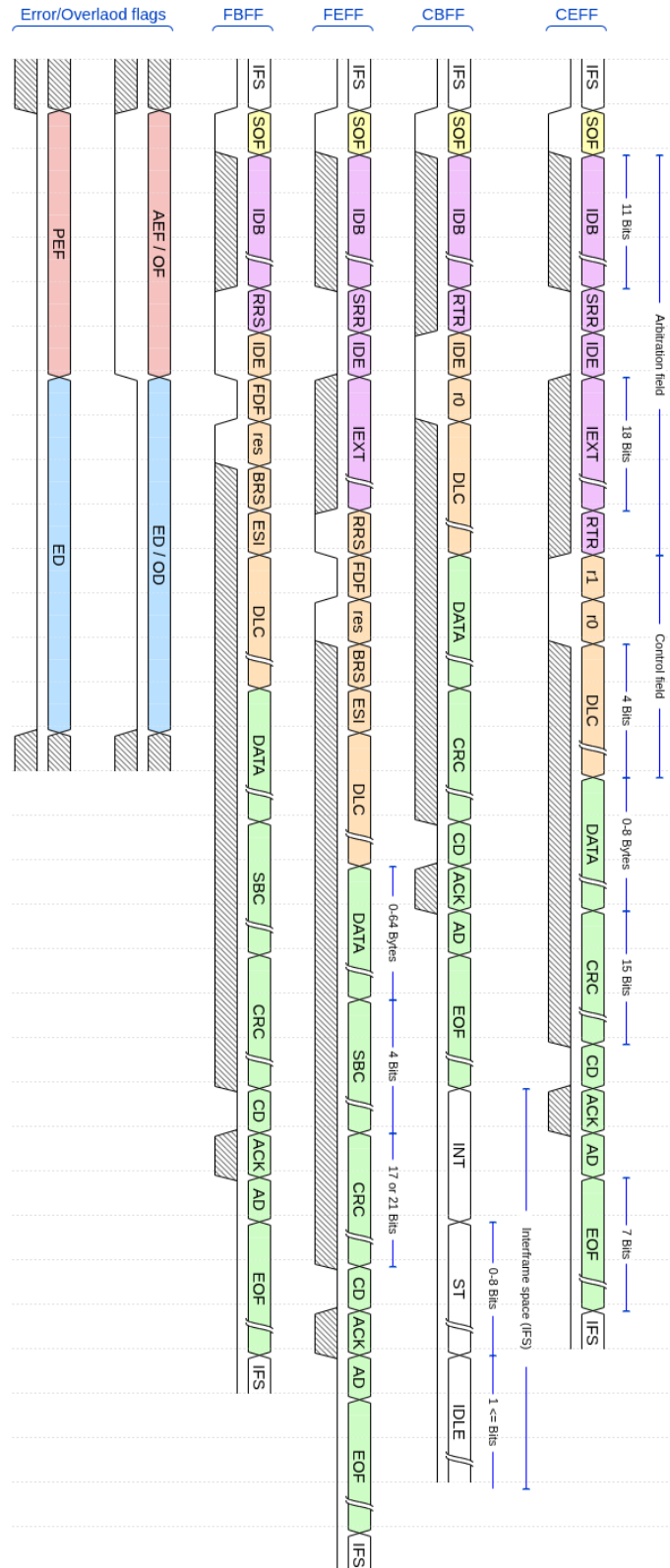


Figure 4: CAN frame formats (CBFF, CEFF, FBFF, FEFF) and the error and overload flags, with field widths annotated per ISO 11898-1. The bus waveform below each format indicates the level of fixed-polarity protocol bits. Hatched regions indicate variable-content fields.

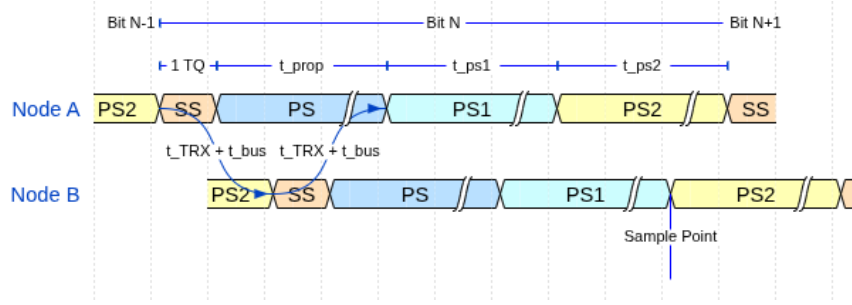


Figure 5: CAN bit time segments (SS, PS, PS1, PS2) and Sample Point (SP). The two-node layout illustrates the round-trip propagation delay ($t_{\text{TRX}} + t_{\text{bus}}$ each way) that PS must cover for correct arbitration.

- **Sync Segment (SS):** one TQ. The segment at which a recessive-to-dominant edge is expected when the node is synchronized.
- **Propagation Segment (PS):** guarantees that a bit driven by any node reaches all others before the sample point, enabling correct arbitration. PS shall be programmed to at least twice the one-way propagation delay: $2 \times (t_{\text{TRX}} + t_{\text{bus}})$, see Figure 5.
- **Phase Segment 1 (PS1):** immediately precedes the sample point. Can be lengthened by the resynchronization mechanism to absorb positive phase errors.
- **Phase Segment 2 (PS2):** follows the sample point to the end of the bit. Can be shortened to absorb negative phase errors.

The bus is sampled at the sample point (SP) which falls at the PS1 / PS2 boundary.

3.3.1 Synchronization

In a synchronized receiving node, edges arrive within SS. An edge outside SS carries Phase Error (PE) and triggers synchronization to realign the sample point at the PS1/PS2 boundary. During the frame, resynchronization on each qualifying recessive-to-dominant edge adjusts the bit time based on the PE relative to SS: a positive PE lengthens PS1 by up to the Synchronization Jump Width (SJW), and a negative PE shortens PS2 by up to SJW. Hard synchronization, triggered by the SOF dominant edge and the FDF-to-res dominant edge, restarts the bit time with SS completed. Only one synchronization is permitted per bit time (REQ-026). Figure 6 illustrates how two consecutive mid-frame synchronization events align an unsynchronized receiving node to the transmitter on the bus.

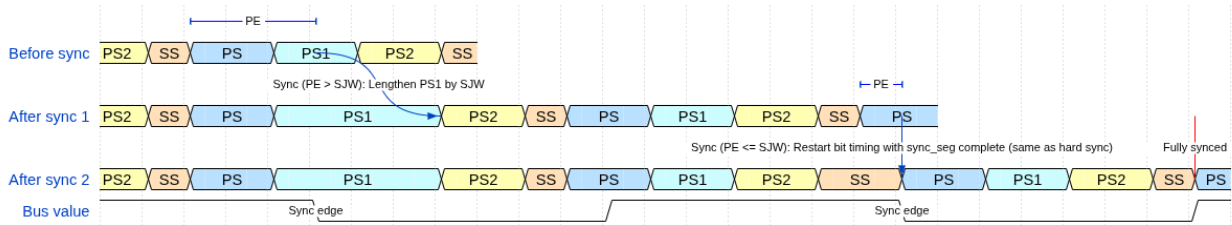


Figure 6: Resynchronization over two successive sync edges. PE is the phase error relative to SS. SJW is the Synchronization Jump Width.

3.3.2 Transmitter Delay Compensation

When the CF data phase begins, the transceiver loopback delay may span multiple data-phase bit times, necessitating a delay before sampling to compensate for in-flight bits (REQ-025). TDC measures this delay in the nominal-rate control field: from when the res bit goes dominant on TX to when the edge arrives on

RX. The Secondary Sample Point (SSP) is then placed at the measured delay plus a programmable offset, firing before the SP in each bit time so that bit errors can be reacted upon at the SP. For the initial data-phase bits where the loopback has not yet returned, the SSP is suppressed. From the first valid SSP onward, SSP monitoring replaces SP monitoring for the remainder of the data phase, see Figure 7.

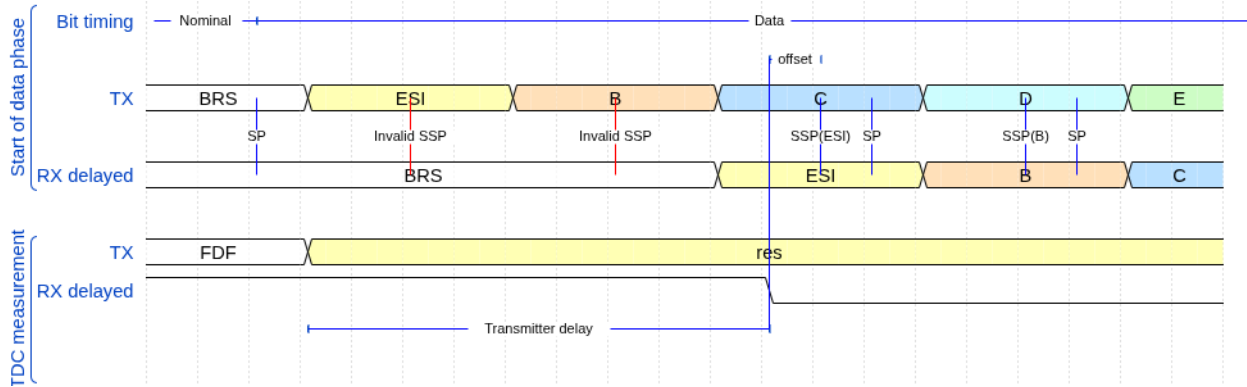


Figure 7: Transmitter Delay Compensation. Top: the first data-phase bits have no valid SSP while the loopback is still in transit. Once the loopback returns, the SSP is placed at the measured transceiver delay plus a programmable offset, firing before the SP in each bit time. Bottom: the transceiver delay is measured from when the res bit goes dominant on TX to when the edge arrives on RX.

3.4 Bit Stuffing

Bit stuffing ensures sufficient transitions on the bus for receiver clock synchronization. CC applies dynamic stuffing from SOF through the CRC field: after five consecutive bits of the same polarity, the transmitter inserts one complement Stuff Bit (SB) and the receiver discards it (REQ-018). CAN FD retains dynamic stuffing through the arbitration and data fields, then switches to fixed stuffing in the CRC field. Fixed Stuff Bits are inserted at fixed positions - before the SBC field and after every fourth CRC bit - each the inverse of the preceding bit, guaranteeing transitions regardless of the CRC data pattern. A separate Stuff Bit Count (SBC) field, Gray-coded with a parity bit, enables receivers to verify the number of dynamic stuff bits inserted in the frame (REQ-016, Figure 8).

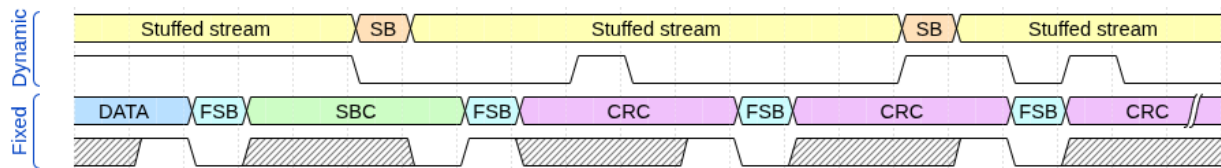


Figure 8: Dynamic and fixed bit-stuffing examples showing stuff bit placement for both encoding modes. The waveform below each row shows the resulting bus signal.

3.5 Cyclic Redundancy Check

The transmitter divides the frame bit stream by a fixed generator polynomial and appends the remainder as the CRC field. Receivers repeat this operation and flag a mismatch as a CRC error. The polynomial and field length depend on frame type and data payload length (REQ-006) as listed below.

- **CRC-15:** Used for all CC frames. The CRC accumulates over SOF, arbitration, control, and data fields with SBs excluded (REQ-013).
- **CRC-17:** Used for FD frames with data payloads up to 16 bytes (DLC 0-10).
- **CRC-21:** Used for FD frames with data payloads from 20 to 64 bytes (DLC 11-15).

The key asymmetry between CC and FD concerns the CRC data feed. For FD frames, dynamic stuff bits up to and including the data field are included in the CRC computation, along with the SBC field itself. Fixed stuff bits are excluded from the CRC computation in both CC and FD frames (REQ-013). The CRC field is terminated by a recessive CRC delimiter bit (REQ-017).

3.6 Error Detection and Fault Confinement

Every CAN node monitors the bus for five categories of error (REQ-021):

- **Bit error:** A transmitter reads back a polarity different from what it drove. Exceptions: a recessive non-stuff bit overridden during arbitration and a recessive bit in the ACK slot.
- **Stuff error:** Six consecutive bits of the same polarity in frame fields where dynamic bit stuffing applies.
- **CRC error:** Received checksum does not match the locally recomputed value.
- **Form error:** A fixed-format field contains an illegal bit value.
- **Acknowledgment error:** A transmitter receives no dominant ACK bit from any receiver (REQ-014).

Detection of any error causes the detecting node to abort frame transmission or reception and start transmitting an error flag (REQ-022). An error active node transmits an Active Error Flag (AEF) of six consecutive dominant bits. An error passive node transmits a Passive Error Flag (PEF) of six consecutive recessive bits instead (REQ-007). Both are followed by an eight-bit recessive error delimiter. Transmission of an AEF flag will trigger bit-stuffing errors in the receiving nodes on the bus, causing these nodes to transmit their own error flags.

The FCE tracks each node's error history through the Transmit Error Counter (TEC) and Receive Error Counter (REC). Counter increments and decrements follow the rules defined in REQ-028. A node begins in error active, transitions to error passive when either counter exceeds 127, and to bus off when TEC exceeds 255 (REQ-029). An error-passive transmitter is exempt from the TEC increment on an ACK error, preventing an isolated node from escalating to bus-off through failed acknowledgments alone. In the bus off state, the node ceases all bus activity and shall not influence the bus (REQ-027) until 128 sequences of 11 consecutive recessive bits are observed, after which TEC and REC are reset and the node returns to error active. A host-initiated reset also returns the FCE to its initial state immediately (REQ-029).

4 Verification Plan

The verification plan adds five classification dimensions to each of the 37 requirements, driving both the module architecture and the testbench design. The plan was populated using MCP introduced in Section 2. The five classification dimensions fall into two groups. Design-facing dimensions inform the module architecture. Verification-facing dimensions inform the verification strategy and testbench design. Each field is summarized in the bullets below and described in detail in the following sections.

- **Design-facing:** `layer`, `side`, and `format_applicability` - determining module ownership, TX/RX path decomposition, and which frame formats each requirement applies to.
- **Verification-facing:** `observability` and `verification_method` - determining whether internal state is required and specifying the verification technique.

4.1 Layer

The `layer` field assigns each requirement to the sub-layer that owns it: LLC, MAC, PCS, or FCE. A fifth label, `system`, marks requirements that are inherently multi-layer or multi-node and need either an integrated testbench or a two-node simulation. [REQ-020](#) (arbitration loss) illustrates the latter: verifying it requires MAC, PCS, and FCE cooperating in a full node, plus a second node driving dominant while the DUT drives recessive.

4.2 Side

The `side` field classifies each requirement as transmitter, receiver, or both, reflecting the ISO standard's practice of specifying TX and RX obligations separately. It scopes the testbench to the corresponding drive mode. [REQ-025](#) (TDC) illustrates a TX-only requirement - measuring transceiver loop delay and positioning the SSP only apply to transmitting nodes.

4.3 Format Applicability

The `format_applicability` field records which of the in-scope frame formats (CBFF, CEFF, FBFF, FEFF) each requirement applies to. Because the formats differ in stuffing mode, CRC polynomial, and control field structure, a requirement that applies only to CF frames implies stimulus with FDF=1 and DLC values covering both sides of the CRC-17/CRC-21 boundary, while one that applies to all four formats requires a configuration for each.

4.4 Observability

The `observability` field classifies each requirement as either black-box or white-box, relative to the module boundary of the owning layer:

- **Black-box:** Can be verified purely through the module's observable port signals.
- **White-box:** Verification requires direct observation of the module's internal state.

This classification drives the testbench architecture. Black-box requirements can be verified by driving the module inputs and checking the output signals. White-box requirements require a parallel reference model or direct observation of internal signals. [REQ-006](#) (CRC polynomial selection) is representative: the polynomial in use - CRC_15, CRC_17, or CRC_21 - is configured inside `can_mac_crc` and never exposed at the module boundary. Verification uses a reference model that independently computes the expected CRC for each format and DLC combination and compares it against the transmitted sequence.

4.5 Verification Method

The `verification_method` field specifies how each requirement will be checked. Four methods are used:

- **simulation:** Assertion procedures in a testbench.
- **code_inspection:** RTL source review.
- **waveform_inspection:** Manual review of simulation output.
- **coverage:** A functional coverage bin confirming a specific condition was exercised.

Combinations are valid when multiple sub-claims within one requirement each call for a different method. [REQ-018](#) (bit stuffing) illustrates this: `simulation` assertions verify that stuff bits are inserted and removed at the correct positions, while `coverage` bins confirm that the edge case of five consecutive identical bits landing at a field boundary was exercised at least once.

4.6 Traceability: Label and File

The `label` and `file` fields establish a direct, navigable link from each requirement to its verification artifact. `file` identifies the testbench or RTL source file responsible for covering the requirement. `label` identifies a specific named procedure, assertion, or coverage ID within that file.

4.7 Status

The `status` field (`not_started`, `in_progress`, `complete`) records requirement closure state explicitly, allowing partial progress to be tracked.

4.8 Verification Plan Summary

Table 5 shows the 37 requirements distributed across layer, side, format scope, and observability. MAC carries 20 of the 37, with 16 white-box, reflecting the breadth of frame-encoding logic that requires bit-level state access to verify. FCE is the opposite: both requirements are black-box, since fault-confinement state transitions are fully observable through the node's error-state output signals. The complete verification plan is reproduced in Section C as two separate tables linked by common IDs.

Table 5: Requirement distribution by layer, side, format scope, and observability. n = total. FS = format-specific (not applicable to all four frame formats). BB = black-box. WB = white-box.

Layer	Requirements	n	TX	RX	Both	FS	BB	WB
MAC	REQ-006 , REQ-008 , REQ-010–013 , REQ-015–019 , REQ-021–023 , REQ-030–032 , REQ-034–035 , REQ-037	20	3	0	17	5	4	16
LLC	REQ-001–005 , REQ-036	6	3	1	2	0	3	3
PCS	REQ-024–027	4	1	0	3	1	1	3
FCE	REQ-028–029	2	0	0	2	0	2	0
System	REQ-007 , REQ-009 , REQ-014 , REQ-020 , REQ-033	5	1	0	4	0	2	3
Total		37	8	1	28	6	12	25

5 Design and Architecture

The design maps each ISO 11898-1 sub-layer to a dedicated module, as shown in Figure 9. The primary data path runs from `can_llc` through `can_mac` to `can_pcs`, with `can_fce` sitting outside the path as a cross-cutting entity. `can_mac` bundles the four MAC submodules into a single sub-layer wrapper. `can_mac_pcs_fce` further combines `can_mac`, `can_pcs`, and `can_fce` into a synthesizable integration target. A centralized types package (`pk_can_types`) defines all protocol constants, interface records, and reset values shared across modules.

- **can_llc:** Implements the LLC sub-layer. Accepts host frames over Avalon-ST, applies acceptance filtering on RX, and streams frames to and from the MAC.

- **can_mac_fsm**: Orchestrates frame TX and RX across all four in-scope formats, controls ready back-pressure on can_mac_ser, drives can_mac_bs and can_mac_crc, and commits the next bit to the can_pcs.
- **can_mac_ser**: Receives the LLC frame over Avalon-ST and presents a serialized bit stream to can_mac_fsm, asserting valid on each bit.
- **can_mac_bs**: Performs dynamic and fixed bit stuffing and destuffing and generates the SBC field for FD frames.
- **can_mac_crc**: Runs CRC-15, CRC-17, and CRC-21 engines in parallel, selecting the output based on frame format and DLC.
- **can_pcs**: Applies bit timing, generates SP and SSP strobes, performs hard synchronization and resynchronization, and manages the TX/RX interface to the transceiver.
- **can_fce**: Tracks transmit and receive error counts, manages error-active/error-passive/bus-off state transitions, and signals bus-off entry and recovery.

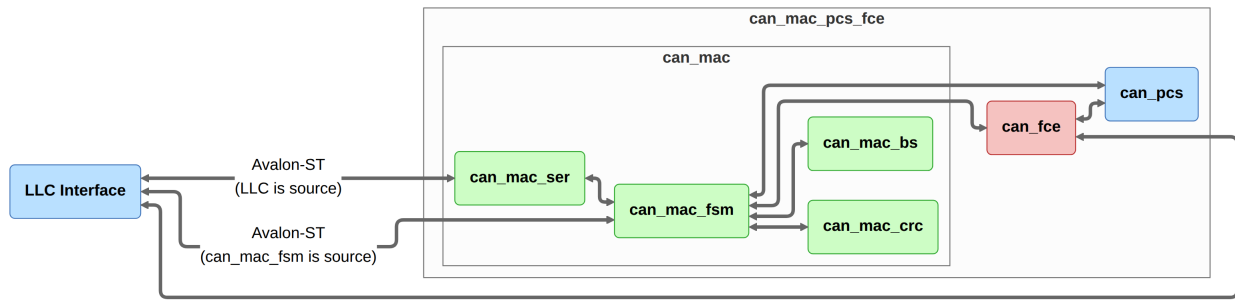


Figure 9: Implementation module decomposition showing the five entities and their inter-module connections.

5.1 Adopting the ISO 11898-1 Sub-layer Model

Mapping each ISO 11898-1 sub-layer to a dedicated module is the natural decomposition. Module boundaries align directly with verification targets, each requirement points unambiguously to the responsible implementation unit, and each module can be exercised in isolation without driving frame-level stimulus through unrelated sub-layers.

5.2 Per-Field FSM Granularity

The `format_applicability` dimension expresses requirements at the level of individual protocol fields. [REQ-037](#) defines the complete MAC frame field sequence and fixed-polarity bits for all in-scope formats. With one FSM state per protocol field, the FSM naturally progresses through that structure: format-dependent transitions - such as the divergence between CC and FD at the FDF field - become state graph edges rather than counter conditionals inside a shared state, and each requirement maps directly to a named FSM state.

5.3 LLC Frame Format

The LLC layer defines two frame formats: a host-facing format that maintains compatibility with `can_bus_controller`, and a MAC-facing format designed to efficiently facilitate frame streaming.

5.3.1 Host-LLC Interface Format

The host-LLC format shown in Figure 10 extends the `can_bus_controller` LLC frame format, expanding the data field from 8 to 64 bytes and adding FDF, BRS, and ESI to the existing trailing control bytes alongside IDE and RTR. The field layout and byte positions are otherwise unchanged, so host software requires no change to the fields it already uses.

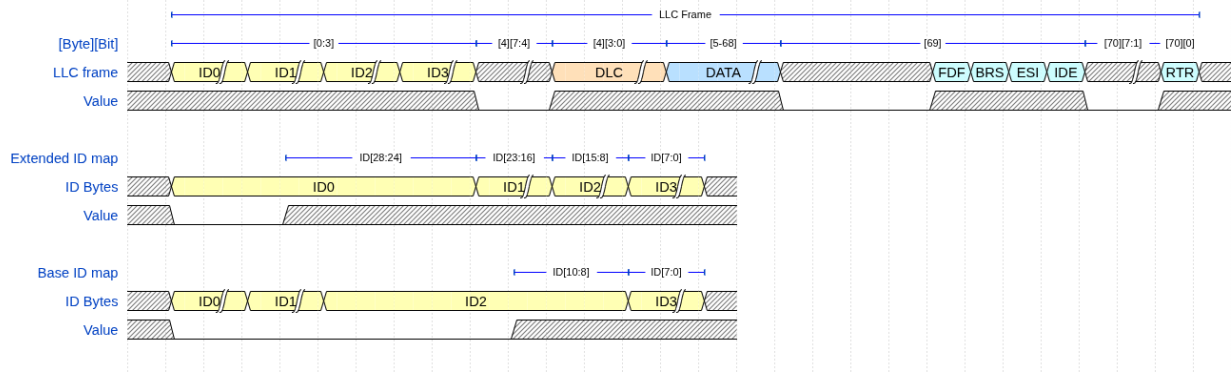


Figure 10: LLC frame format (71 bytes) at the host-LLC interface, with identifier byte mapping for base and extended IDs. Hatched regions indicate variable-value bits.

5.3.2 LLC-MAC Interface Format

The host-LLC format places all control flags after the payload, requiring a full 71-byte frame buffer before serialization can begin. The LLC-MAC format (Figure 11) avoids this by packing all metadata into two leading config bytes: `can_mac_ser` receives these first, then streams the ID and data bytes immediately.

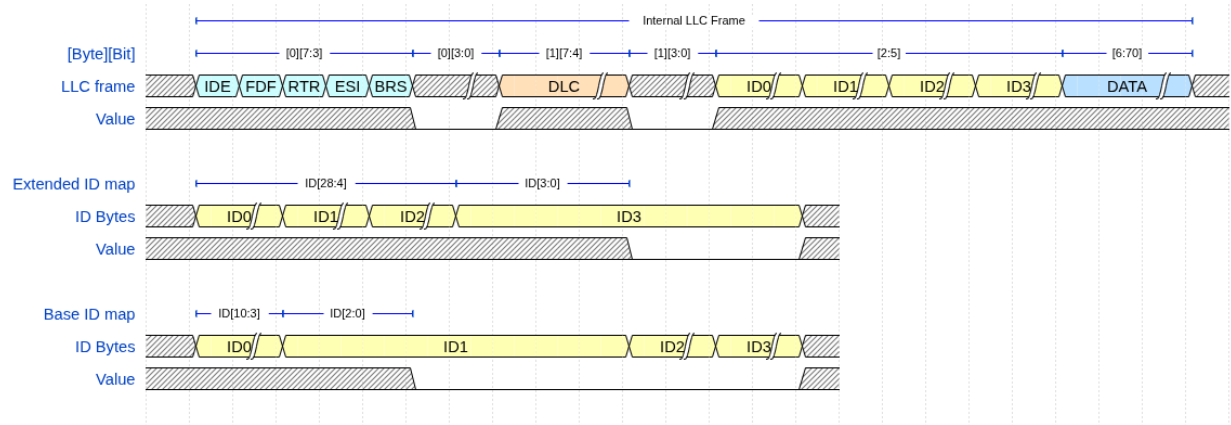


Figure 11: Internal LLC frame format at the `can_mac_ser` input, with identifier byte mapping for base and extended IDs. Hatched regions indicate variable-value bits.

6 Implementation

All inter-module interfaces use typed records (e.g., `t_can_mac_pcs_if_m2s`, `t_can_mac_fsm_bs_if_s2m`), each paired with a reset constant (e.g., `c_can_mac_pcs_if_m2s_reset`) so every module can be reset without enumerating individual fields. Port direction follows `m2s/s2m` (master-to-slave/slave-to-master) for control interfaces and `s2d/d2s` (source-to-destination/destination-to-source) for Avalon-ST data interfaces. `pk_can_types` is the single shared package all modules depend on. It defines every interface record type, protocol constant, frame format byte layout, and utility function used across the design.

can_llc was not implemented within the project schedule. Its interface contracts are fully specified in the verification plan (REQ-001 through REQ-005, REQ-031, REQ-034, REQ-036), and the implementation path is described in Section 9.2.

6.1 Module Overview

can_mac_pcs_fce is the synthesis and integration boundary. It instantiates can_mac, can_pcs, and can_fce. can_mac in turn instantiates can_mac_fsm, can_mac_ser, can_mac_bs, and can_mac_crc, with can_mac_fsm being the central orchestrating FSM in the MAC layer.

6.2 can_mac_fsm

can_mac_fsm accumulates received frames into a 70-byte internal byte array (llc_frame), reusing can_bus_controller's register-array approach as a proof-of-concept baseline. The resource overhead is negligible at 15 bytes but becomes the dominant logic element cost at 70 bytes (Section 8.1). A block RAM migration is the identified upgrade path (Section 9.2). Bus-off state is owned entirely by can_fce. can_mac_fsm treats fce_i.bus_off as a secondary reset alongside the hardware reset, suppressing all bus activity. The complete signal-level interface is reproduced in Section B.

6.2.1 FSM Structure and Mode Flag

can_mac_fsm contains two synchronous processes (p_fsm and p_stream_to_LLC). An is_transmitter flag, latched when the FSM drives the SOF dominant bit at the start of a new frame transmission and cleared at arbitration loss (REQ-020) or at the end of the EOF field, partitions per-state logic into a TX branch and an RX branch. The error-frame states (s_error_flag, s_error_delimiter) are an exception: both transmitter and receiver errors enter the same two-state sequence, with flag polarity driven by fce_i.error_active (REQ-007, REQ-022).

p_fsm organizes each sample-point cycle as three phases:

- **Pre-case:** handles all conditions that preempt normal state progression - lost arbitration (REQ-020), bit errors (REQ-021), and ACK errors (REQ-014) on the TX branch, and stuff errors (REQ-018), form errors (REQ-021), overload conditions (REQ-023), and deferred SBC/CRC mismatch detection (REQ-016, REQ-021) on the RX branch. When any of these fire, v_skip_case is set and the case block is bypassed entirely.
- **Case:** handles only the error-free state transitions - bit_count advancing and state changing according to the protocol field sequence.
- **Post-case:** feeds the BS and CRC engines at the sample point and commits the drive polarity to the PCS output.

The structure trades per-state locality for non-duplication of cross-cutting logic. Placing stuff-bit handling inside each state would duplicate identical detection and feed logic across the seven dynamic-stuffing states (s_arbitration through s_data). Centralizing it in the pre-case handles it once, and v_skip_case provides a single auditable guarantee that the state machine does not advance when it should not. For logic that is genuinely per-state - DLC parsing, ACK success latching, EOF completion - the case block handles it directly. The locality cost is reasonable since the exception logic is cross-cutting.

p_fsm implements the per-field state granularity introduced in Section 5.2. Each post-arbitration field has a dedicated state, with the arbitration region sharing s_arbitration across ID bits, RTR/SRR/RRS, and IDE via bit_count. The complete FSM is shown in Figure 12.

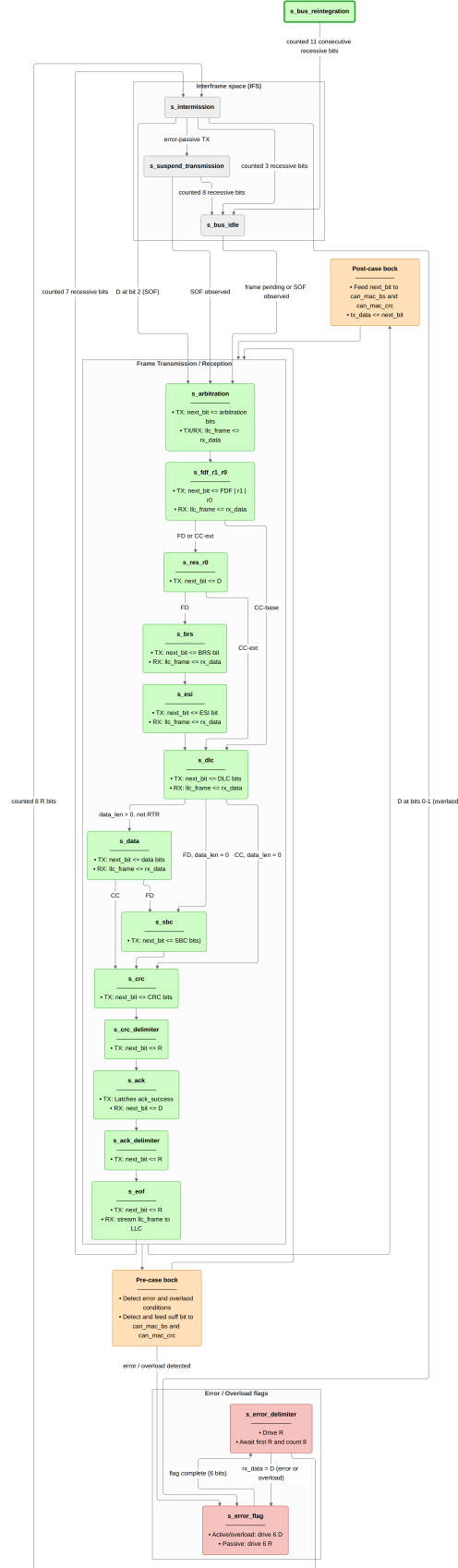


Figure 12: `can_mac_fsm`. FSM orchestrating frame transmission and reception in `can_mac`. The state machine encodes the four in-scope frame formats depicted in Figure 4. Green states encode error-free and stuff-bit-free frame progression. Error and overload detection is handled, and `can_mac_bs` and `can_mac_crc` are fed, in the pre-case code block. The next bit is driven to `can_pcs` and fed to `can_mac_bs` and `can_mac_crc` in the post-case code block.

6.2.2 TX Mode: Frame Transmission

When `is_transmitter = true`, the FSM writes `pcs_o.tx_data` two clock cycles after each sample point, giving the BS and CRC engines time to present valid outputs. `bs_i.valid` must be stable before the TX branch decides whether a stuff bit is due (REQ-018), and `crc_i.crc` must hold the fully accumulated value before the first CRC bit is driven (REQ-006). The PCS latches `pcs_o.tx_data` at the bit boundary - the MAC simply needs valid data ready in time.

The FSM samples the bus at each sample point to detect bit errors (REQ-021), ACK (REQ-014), and arbitration loss (REQ-020). In the CAN FD data phase, the SSP strobe replaces the SP for bit-error monitoring (REQ-025). The MAC compares `pcs_i.rx_data` against `transmitted_bits_shift_reg(tdc_delay)` - where `tdc_delay` is supplied alongside the SSP strobe by the PCS - to check the bit that was transmitted `tdc_delay` bit times earlier. Arbitration loss clears `is_transmitter` in `s_arbitration` and the node continues as an receiver.

During `s_arbitration` multiple nodes may be transmitting, so both TX and RX feed `pcs_i.rx_data` into the CRC and BS engines - the bus is the only authoritative source (REQ-013, REQ-020). A transmitter that loses arbitration therefore needs no handoff: its accumulators already match a pure receiver's. From `s_fdf_r1_r0` onward the transmitter switches to `transmitted_bits_shift_reg(tdc_delay)`, enabling TDC in the data-phase.

6.2.3 RX Mode: Frame Reception

When `is_transmitter = false`, the FSM observes `pcs_i.rx_data` at each sample point and stores received bits directly into the `llc_frame` byte array. The FSM drives the BS engine from `pcs_i.rx_data` to perform destuffing (REQ-018), and the CRC engine accumulates the received bit stream in parallel (REQ-013). The FSM validates the SBC field (FD frames, REQ-016), compares the received CRC against the locally accumulated result (REQ-006), and checks form bits (reserved bits, CRC delimiter, ACK delimiter, EOF) for required polarities (REQ-021). A mismatch in any of these fields triggers a transition to the error-frame sequence (REQ-022). During the ACK slot the FSM drives dominant for one bit (`bit_count = 0`) regardless of frame format (REQ-014). The FD ACK slot spans two bits but the receiver asserts dominant only during the first (REQ-014).

During `s_intermission`, the completed frame is streamed byte-by-byte to the LLC RX sink over the Avalon-ST interface by `p_stream_to_LLC`, a dedicated process running concurrently with `p_fsm`.

6.3 can_mac_ser

`can_mac_ser` converts the LLC byte stream into a serial bit stream for the MAC FSM. Its four-state FSM manages the two-byte configuration handshake, byte fetching, and bit-by-bit serialization. The serializer extracts LLC metadata (IDE, FDF, DLC, FTYP, BRS, ESI) from the two config bytes and registers it in `t_llc_metadata`, which remains stable for the entire frame. The serializer forwards `transfer_status` from `can_mac_fsm` back to the LLC, returning to `s_load_config_byte_0` on any non-ongoing status so that errors and aborts terminate serialization immediately. The four-state serializer FSM is shown in Figure 13.

The 32-bit ID field in the internal format is left-aligned: a base identifier (11 bits) occupies bits [31:21], leaving 21 unused padding bits at the LSB end. An extended identifier (29 bits) occupies bits [31:3], leaving 3 unused padding bits. The serializer tracks this with two counters initialized in `s_load_config_byte_1` from the `ide` flag: `id_bits_remaining` counts real ID bits still to be presented,

and `padding_bits_remaining` counts trailing unused bits to be skipped. In `s_shift_out_bits`, padding bits are advanced without asserting `valid`, so the MAC FSM never observes them.

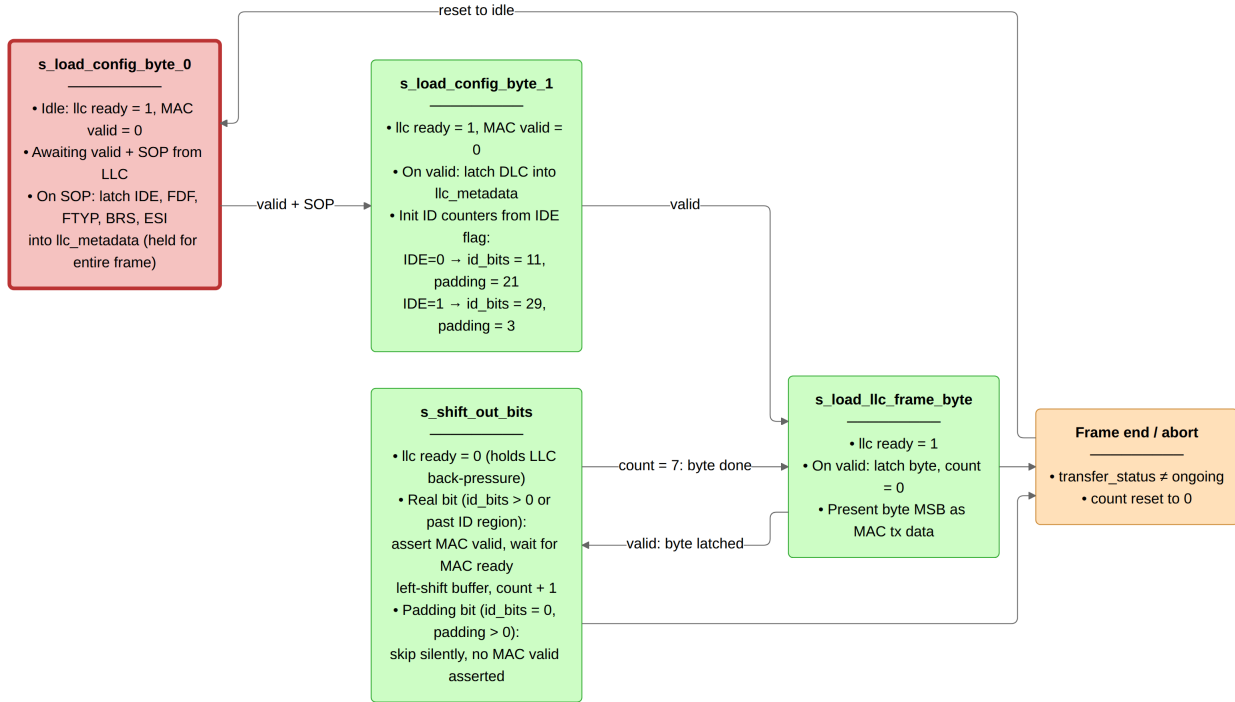


Figure 13: `can_mac_ser` FSM (four states) serializing the internal LLC frame to the MAC bit stream. Unused padding bits in the 32-bit ID field are skipped silently.

6.4 `can_mac_bs`

`can_mac_bs` implements both dynamic and fixed bit stuffing for CC and FD frames (REQ-018). The entity is instantiated inside `can_mac` and serves both TX stuffing and RX destuffing via the same logic: `can_mac_fsm` drives the `can_mac_bs` interface, and the stuffer's output is either inserted into the TX bit stream or used by `can_mac_fsm` to discard SBs and FSBs from the received stream.

In dynamic mode (`fixed_bit_stuffing_en = '0'`), the stuffer counts consecutive bits of identical polarity and emits an inverse-polarity SB after every five. A counter is incremented on each dynamic SB. Its value is Gray-coded and parity bit is added, generating the `stuff_bit_count` output. This value is read by `can_mac_fsm` when transmitting the SBC field (REQ-016).

In fixed mode (`fixed_bit_stuffing_en = '1'`), used for the FD CRC region, a FSB is emitted immediately on the rising edge of `fixed_bit_stuffing_en`, then one FSB every four real bits. If a dynamic SB is already pending when `fixed_bit_stuffing_en` rises, the initial FSB emission is skipped - preventing double stuffing.

6.5 `can_mac_crc`

`can_mac_crc` runs three parallel `gen_crc` instances on separate data feeds and selects the result with an output multiplexer, as shown in Figure 15: `data_cc` drives CRC-15 via `valid_cc`, while `data_fd` drives both CRC-17 and CRC-21 via `valid_fd`. The multiplexer selects the active engine's result based on `crc_poly_select` and left-aligns it to the common 21-bit output width: CRC-15 occupies bits [20:6], CRC-17 occupies bits [20:4], and CRC-21 occupies the full width. Transmitter nodes set `crc_poly_select` from

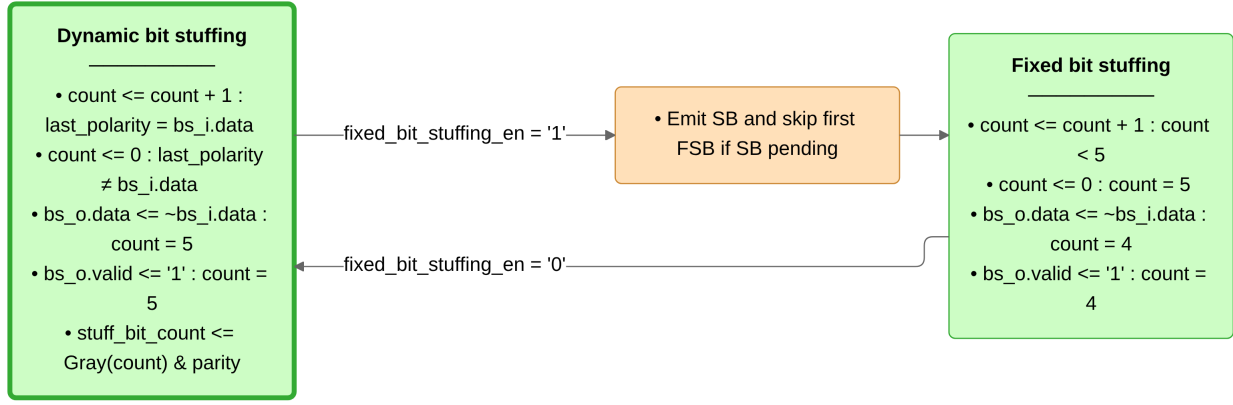


Figure 14: `can_mac_bs` bit stuffing logic.

the DLC field in `llc_metadata` before the first frame bit is driven. Receiving nodes set `crc_poly_select` after the DLC field has been received.

The output mux is kept combinatorial - each `gen_crc` instance registers its outputs, so the mux already selects over registered values, keeping IPT at 2 system clocks. A registered mux would require 3 system clocks, violating [REQ-024](#) ($\text{IPT} \leq 2 t_q$) at minimum prescaler ($m=1, t_q = 1$ system clock).

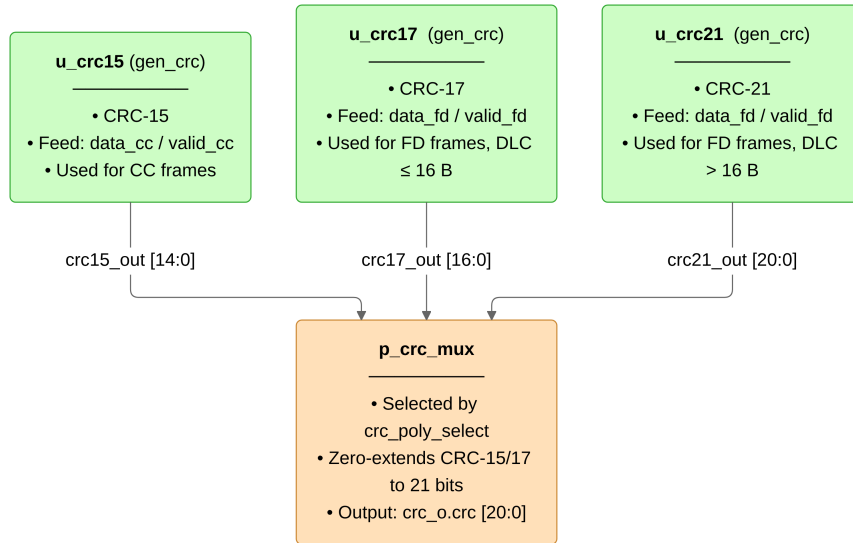


Figure 15: `can_mac_crc` dataflow with three parallel CRC engines. The output mux is combinatorial to satisfy the ISO $\text{IPT} \leq 2 t_q$ constraint at minimum prescaler.

6.6 `can_fce`

`can_fce` implements the error state and counter management specified in [REQ-029](#). It maintains TEC and REC and transitions between three states: `s_error_active`, `s_error_passive` (TEC or REC > 127), and `s_bus_off` (TEC > 255), as shown in Figure 16.

Counter updates follow [REQ-028](#). Bus-off recovery requires 128 separate `pcs_i.idle_condition` pulses, after which both counters reset to zero and the FSM returns to `s_error_active`. `llc_i.normal_mode` resets both counters and returns the FSM to `s_error_active` from any state ([REQ-029](#)).

`can_mac_fsm` asserts `mac_i.passive_tx_ack_error_exempt_1` when it detects an ACK error while the

node is error passive and transmitting, signaling `can_fce` to suppress the TEC increment in accordance with [REQ-028](#).

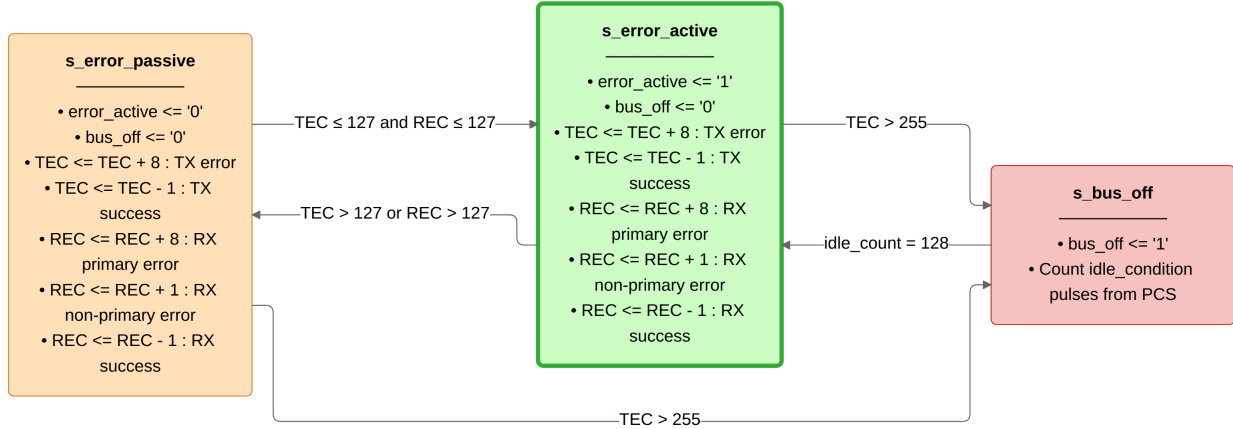


Figure 16: `can_fce` FSM governing the error active, error passive, and bus off node states.

6.7 can_pcs

`can_pcs` implements bit timing as a 4-state FSM and a concurrent TDC pipeline ([REQ-026](#), [REQ-027](#)), see Figure 17. The FSM advances through `s_sync_seg` (1 TQ, fixed), `s_prop_seg`, `s_phase_seg1`, and `s_phase_seg2` as each segment's TQ count expires. The SP strobe and `rx_data` latch fire at the end of `s_phase_seg1`. The TX bit is driven at the end of `s_phase_seg2`. TDC measurement runs at TQ granularity above the segment case statement. When `fce_i.bus_off` is asserted, consecutive recessive bits are counted and `fce_o.idle_condition` is pulsed every 11 bits - enabling FCE bus-off recovery ([REQ-029](#)).

6.7.1 Synchronization

- **Hard synchronization:** Controlled by `can_mac_fsm` via `do_hard_sync`, which restarts the bit time at the sync segment boundary. Used at SOF and at the FDF-to-res transition.
- **Resynchronization:** The `sync_applied` flag enforces one sync per bit time. `rx_bus_prev` gates sync to recessive → dominant edges. The transmitting signal from `can_mac_fsm` suppresses sync during TX. Phase_Seg1 or Phase_Seg2 is adjusted by up to SJW. Phase errors not exceeding SJW produce the same result as a hard synchronization.

6.7.2 Dual Bit Rate Switching

`can_pcs` holds no frame-format knowledge. Rate switching is entirely driven by `can_mac_fsm` through three dedicated control signals.

- `next_bit_is_brs`: At the BRS sample point, `can_pcs` reads BRS polarity and, if recessive, replaces the active segment lengths and SJW with their data-phase counterparts.
- `next_bit_is_res`: Arms TDC measurement at the FD res bit boundary.
- `data_phase_stop`: Asserted by the `can_mac_fsm` at the CRC delimiter SP or on error-frame entry. Restores nominal timing and clears all TDC state.

6.7.3 Transmitter Delay Compensation

TDC is implemented as a three-stage pipeline:

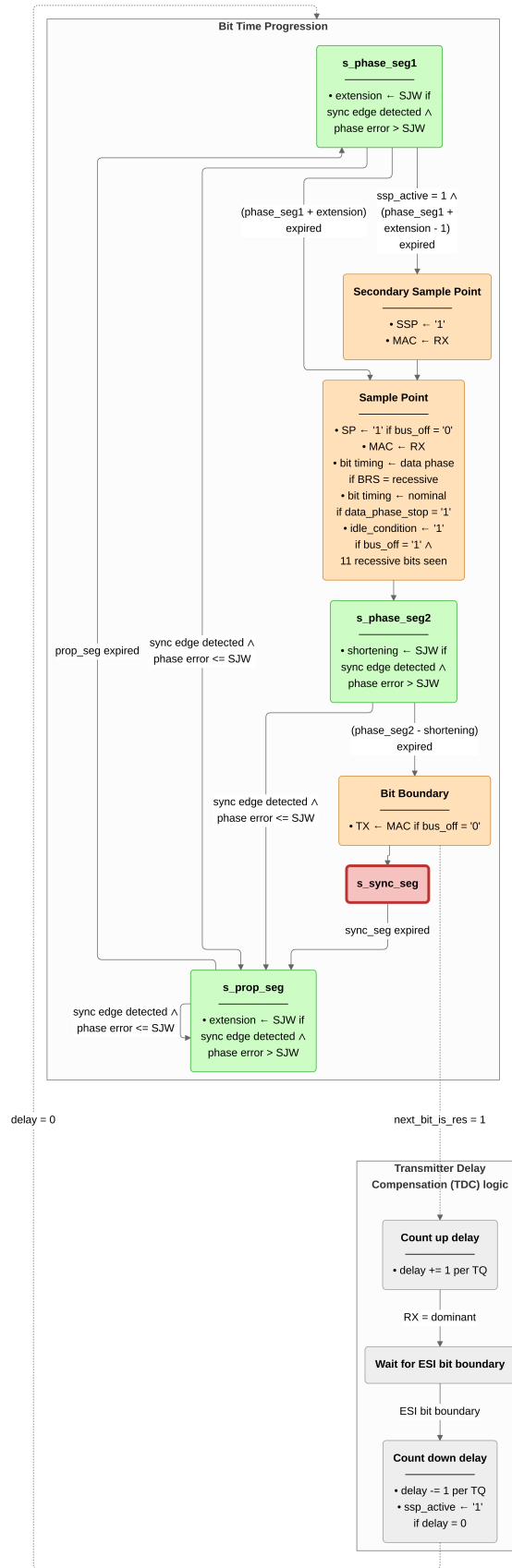


Figure 17: `can_pcs` bit-time FSM with concurrent resynchronization and TDC pipelines per ISO 11898-1 sec. 7.2-7.4.

1. **Measure:** From the FD res bit boundary, `delay_count_tq` increments each TQ until the TX-to-RX dominant edge arrives on RX.
2. **Count down:** At the first data-phase bit boundary, the measured delay is counted down each TQ until zero, setting `ssp_active`.
3. **Fire:** With `ssp_active` set, the SSP fires one TQ before the SP on every data-phase bit time. `can_mac_fsm` uses `tdc_delay` to index into the transmitted-bit shift register for bit error detection.

7 Verification and Results

The implementation described in Section 6 was exercised against the 37-requirement verification plan (Section 4) using five unit testbenches and one integration testbench, `can_mac_pcs_fce_tb`. Figure 18 shows the integration testbench architecture.

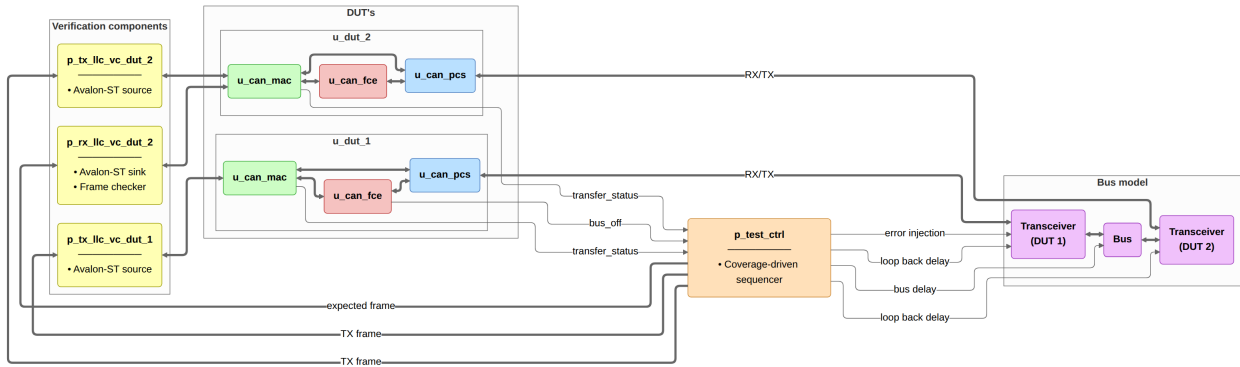


Figure 18: `can_mac_pcs_fce_tb` integration testbench. Two `can_mac_pcs_fce` instances connect through a dominant-wins bus model. Avalon-ST VCs drive and sample the MAC interfaces. `p_test_ctrl` sequences test stimuli, injects bit errors, reads transfer status, and monitors bus-off status.

7.1 Code Inspection

REQ-013 (CRC data feed differs by format: CC excludes stuff bits, FD includes dynamic stuff bits and the SBC field) is verified by code inspection against `can_mac_fsm.vhd` - the FSM controls which frame fields are gated into the CRC engines, and the per-field feed logic is directly readable from the state transitions. **REQ-023** (overload frame conditions) requires triggering an overload frame by injecting a dominant bit at specific field boundaries (last EOF bit, first two intermission bits, last error/overload delimiter bit), which requires a frame-aware bit injector not present in the current testbench.

Code inspection also provides evidence for sub-claims in several requirements covered in Section 7.3:

- **Error detection paths not exercised by simulation:** **REQ-021** sub-claims 2-5 (stuff, form, CRC, and ACK error detection) and **REQ-022** sub-claims 2-3 (data-phase error rate switching and CRC error receiver behavior) are verified by code inspection only - frame-aware error injection is not available in the current testbench.
- **Receiver acceptance of non-standard bit values:** **REQ-012** sub-claims 1-2 (receiver acceptance of dominant SRR and RRS bits without form error) and **REQ-017** sub-claim 1 (two-bit CRC delimiter tolerance) are confirmed by inspecting the form-error logic in `can_mac_fsm.vhd`.
- **Conditions not covered by current testbench stimulus:** **REQ-026** sub-claim 3 (one synchronization per bit time, enforced by `sync_applied`) and **REQ-010** sub-claim 2 (third-intermission-bit skip-SOF transition) are confirmed by reading the guard conditions in `can_mac_fsm.vhd` and `can_pcs.vhd`.

7.2 Unit Testbench Simulation

The five unit testbenches target individual submodules with focused stimulus.

1. **can_mac_crc_tb**: closes [REQ-006](#) via three coverage bins (CRC-15 for CB/CE, CRC-17 for FD ≤ 16 bytes, CRC-21 for FD > 16 bytes). `f_calc_can_crc` computes expected CRC per polynomial and checks DUT output on every frame. Additional checkers verify init vector reset and output stability between frames.
2. **can_mac_bs_tb**: closes [REQ-016](#) and [REQ-018](#) via a two-phase stimulus: six directed FSB tests followed by random dynamic bits. Three concurrent reference models verify DUT output: `p_stuff_bit_checker` verifies a complement bit after every five same-polarity bits, `p_sbc_checker` verifies Gray-code parity and increments on dynamic stuff bits and holds on FSBs, and `p_fsb_checker` verifies initial and periodic FSB polarity. All coverage bins are hit.
3. **can_pcs_tb**: provides simulation evidence for [REQ-024](#), [REQ-025](#), [REQ-026](#) (combined with code inspection), and [REQ-027](#). Two `can_pcs` instances run on mismatched clocks through a physical bus model. Three tests cover reset, random FD frames with alternating clock leadership, and bus-off isolation. `p_check_tdc_delay` verifies at each SSP that `polarity_history(tdc_delay)` matches the sampled bus value. `p_rx_mac_vc` compares RX-sampled bits against the TX sequence bit by bit.
4. **can_fce_tb**: closes [REQ-028](#), [REQ-029](#), and [REQ-033](#) (sub-claim 2) using directed stimulus only - counter rules are deterministic. Reset confirms outputs clear and `llc_i.normal_mode` restores error-active from any state. All [REQ-028](#) counter rules are exercised at the error-active/error-passive boundary. [REQ-033](#) sub-claim 2 confirms no bus-off entry when `passive_tx_ack_error_exempt_1` is set while error-passive. [REQ-029](#) boundary check: 64 `idle_condition` strobes confirm no premature release, 128 confirm clearance and error-active restoration.
5. **can_mac_ser_tb**: closes [REQ-031](#) (DLC encoding: linear for DLC 0-8, FD non-linear for DLC 9-15) and provides supporting evidence for higher-level requirements. Three coverage dimensions drive frame selection: IDE (base/extended), FDF (CC/FD), and DLC (0-15). `p_mac_fsm_vc` verifies all metadata fields against the LLC frame and compares the output bit stream byte by byte, accounting for ID width, padding, and DLC-derived data length. Random back pressure and a 2% mid-frame abort rate exercise pipeline stalls and the `c_disturbed` path. `p_transfer_status_checker` monitors `transfer_status` every cycle.

7.3 Integration Testbench Simulation

`can_mac_pcs_fce_tb` is the primary integration testbench, exercising two `can_mac_pcs_fce` instances connected through a dominant-wins bus model and covering 17 requirements spanning MAC frame encoding, arbitration, and error handling. The following waveforms are selected excerpts from the test run. Many requirements are verified by waveform inspection across all test scenarios. Reproducing a dedicated figure for each requirement would be impractical, so only representative scenarios are shown.

- **Complete FD Frame Transmission** (Figure 19): a complete FD frame transmitted by DUT 1 and received by DUT 2, showing hard synchronization on SOF, TDC loopback delay measurement, dual bit rate switching at the BRS sample point, SSP pulses during the data phase, and ACK confirmation ([REQ-010](#), [REQ-012](#), [REQ-014](#), [REQ-015](#), [REQ-017](#), [REQ-026](#)).
- **Bit Stuffing** (Figure 20): dynamic and fixed mode behavior ([REQ-016](#), [REQ-018](#)).
- **Bit Rate Switching and TDC** (Figure 21): the PCS switches to data-phase bit timing at the BRS sample point and positions the SSP once the transceiver loopback delay is measured ([REQ-025](#), [REQ-030](#)).
- **Arbitration** (Figure 22): the losing node clears `is_transmitter` in-place at `s_arbitration` and

continues as receiver without a state transition (REQ-020).

- **Error Handling and Bus-off Recovery** (Figure 23, Figure 24): error frame escalation and bus-off recovery (REQ-007, REQ-008, REQ-009, REQ-021, REQ-022, REQ-028, REQ-029, REQ-033 sub-claim 1).

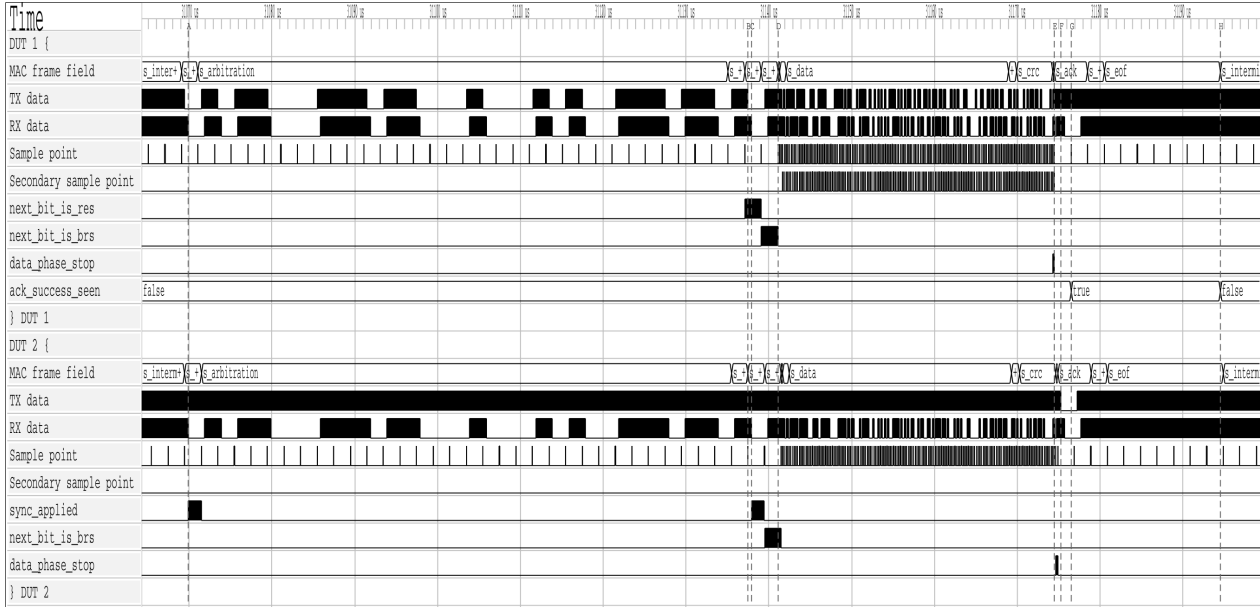


Figure 19: Two-node simulation of a complete FD frame transmission in can_mac_pcs_fce_tb. DUT 2 hard-synchronizes on SOF at A. DUT 1 measures the TDC loopback delay between B and C. Both nodes switch to data-phase bit timing at the BRS sample point at D. DUT 1 switches back to nominal bit timing at the CRC delimiter sample point at E. DUT 2 drives the ACK slot dominant at F, DUT 1 samples the dominant ACK at G and latches ack_success_seen. Transmission ends at H.

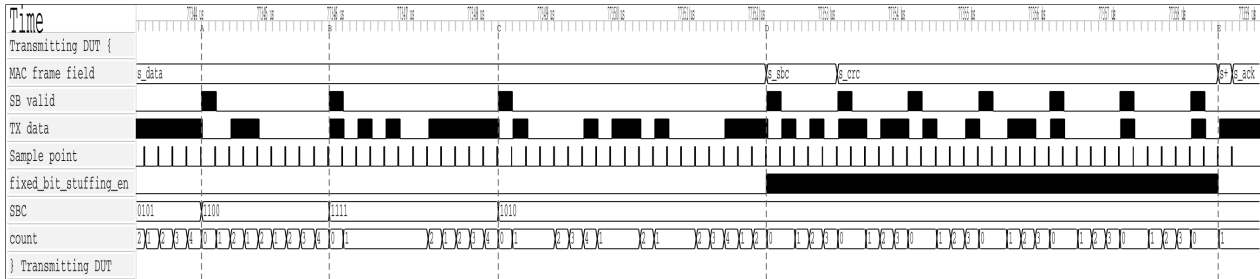


Figure 20: Dynamic and fixed bit stuffing in can_mac_pcs_fce_tb. Dynamic stuff bits are inserted at A, B, and C in the s_data region. At D, fixed_bit_stuffing_en asserts and the stuffer switches to fixed mode for s_sbc and s_crc. Seven fixed stuff bits are inserted between D and E.

7.4 Testbench Results Summary

Of the 37 requirements, 28 are closed: 26 via testbench simulation (Table 6) and two (REQ-013, REQ-023) via code inspection (Section 7.1). Nine remain open: six are LLC requirements (REQ-001 through REQ-005, REQ-036) deferred pending can_llc implementation, REQ-034 (P3) and REQ-035 (P2) are non-blocking, and REQ-021 (P1) is partially covered - bit-error detection is verified in test_bus_off, while the remaining sub-claims (stuff, form, CRC, ACK) require frame-aware error injection not available in the current testbench (Section 9.2).

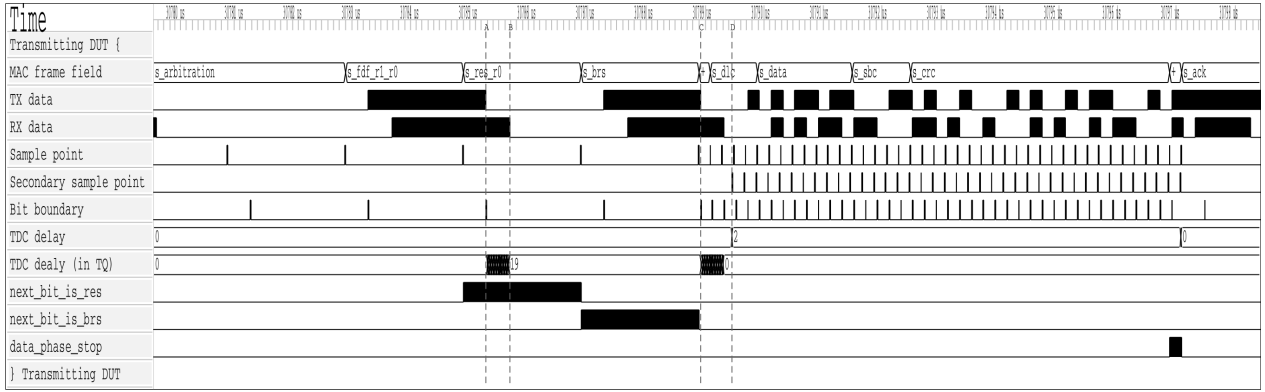


Figure 21: Dual bit rate switching and TDC measurement in can_mac_pcs_fce_tb. At A, the PCS begins counting the transceiver loopback delay in TQ increments. At B, the transmitted bit arrives on RX and the count stops at 19 TQ. At C, the first data-phase bit (ESI) is transmitted and the measured delay is counted down. When the countdown terminates at D, the SSP strobe activates and the TDC delay of 2 TQ is reported to the MAC. next_bit_is_res and next_bit_is_brs mark the measurement window boundaries. data_phase_stop signals the end of the data phase.

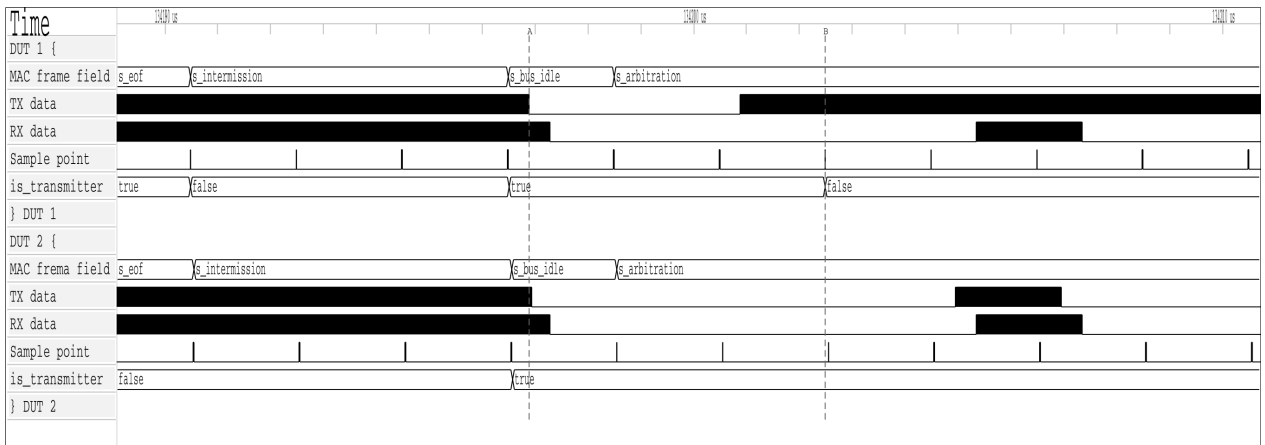


Figure 22: Arbitration loss in can_mac_pcs_fce_tb. Both nodes enter s_arbitration as transmitters at A. DUT 1 loses arbitration at B after transmitting recessive and sampling dominant, and continues as receiver with is_transmitter cleared.

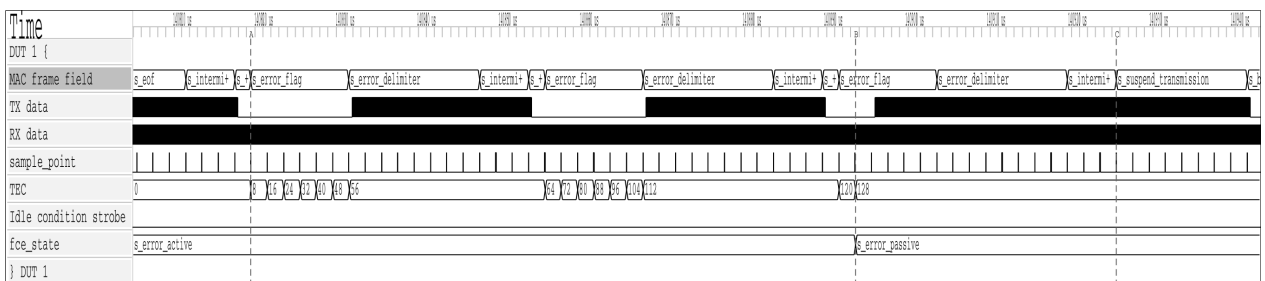


Figure 23: Error frame escalation in can_mac_pcs_fce_tb. A bit error on the SOF bit triggers the first error flag at A, incrementing TEC to 8. Each dominant bit of the error flag is sampled as a bit error because the bus is held recessive, rapidly escalating TEC. At B, TEC reaches 128 and fce_state transitions to s_error_passive. At C, the node enters s_suspend_transmission after s_intermission.



Figure 24: Bus-off recovery in `can_mac_pcs_fce_tb`. DUT 1 transmits the first active error flag at A. At B, TEC reaches 128 and the node transitions to error passive. At C, TEC reaches 256 and the node enters bus off. The FCE counts 128 idle condition strobes from the PCS and restores `s_error_active` at D. At E and F, DUT 2 acknowledges the first two frames transmitted after recovery.

Table 6: Testbench execution status and requirements coverage.

Testbench	Requirements covered	Status
<code>can_mac_crc_tb</code>	REQ-006	Pass
<code>can_mac_bs_tb</code>	REQ-016 , REQ-018	Pass
<code>can_pcs_tb</code>	REQ-024 , REQ-025 , REQ-026 , REQ-027	Pass
<code>can_fce_tb</code>	REQ-028 , REQ-029 , REQ-033 (sub-claim 2)	Pass
<code>can_mac_ser_tb</code>	REQ-031	Pass
<code>can_mac_pcs_fce_tb</code>	REQ-007 , REQ-008 , REQ-009 , REQ-010 , REQ-011 , REQ-012 , REQ-014 , REQ-015 , REQ-017 , REQ-019 , REQ-020 , REQ-021 (bit error only), REQ-022 , REQ-030 , REQ-031 , REQ-032 , REQ-033 (sub-claim 1)	Pass

8 Synthesis

The implemented `can_mac_pcs_fce` stack was synthesized targeting the Cyclone 10 LP device used in Everllence’s IO-extender board. The clock constraint was set to 6 ns (166 MHz), deliberately overconstraining relative to any realistic CAN FD system clock to expose worst-case timing paths.

8.1 Resource Utilization

The synthesized design (`can_mac_pcs_fce`) uses 4,608 Logic Elements (LE) (30% of the target device) and 869 dedicated registers. Table 7 shows the per-module resource breakdown.

Table 7: Resource utilization on Cyclone 10 LP: `can_mac_pcs_fce` breakdown and `can_bus_controller` baseline.

Module	LEs	Registers
<code>can_mac_fsm</code>	4,109	684
<code>can_pcs</code>	190	49
<code>can_fce</code>	117	31
<code>can_mac_crc</code>	84	53
<code>can_mac_ser</code>	84	40
<code>can_mac_bs</code>	31	12

Module	LEs	Registers
CAN FD total (can_mac_pcs_fce)	4,608	869
CC (can_bus_controller)	1,146	334 Existing controller, see Section 1.1.3

can_mac_fsm dominates at 89% of total LEs, driven primarily by the RX frame buffer. can_mac_fsm accumulates received frames into a 70-byte (560-bit) internal byte array, and each buffer register drives combinatorial decode and mux logic. The five remaining modules together consume 506 LEs, modest overhead relative to the frame buffer.

8.2 Timing Results

Timing analysis covers four PVT corners: slow/fast transistor speeds, 1150 mV supply voltage, and -40°C to 100°C temperature range. Table 8 shows the setup and hold slack at the overconstraining 166 MHz clock.

Table 8: Timing results for can_mac_pcs_fce at 6 ns (166 MHz) on Cyclone 10 LP. Slow/Fast denotes process transistor speed.

Corner	Worst setup slack	fmax estimate	Hold slack
Slow 1150 mV 100°C	-1.858 ns	~127 MHz	+0.236 ns
Slow 1150 mV -40°C	-1.550 ns	~152 MHz	+0.021 ns
Fast 1150 mV 100°C	+0.230 ns	>166 MHz	+0.163 ns
Fast 1150 mV -40°C	+0.580 ns	>166 MHz	+0.145 ns

The worst-case fmax is approximately 127 MHz on the slow 100°C corner. The negative setup slack on slow corners indicates timing violations at the 166 MHz constraint - a consequence of the deliberate overconstrain. Practical CAN FD implementations use system clocks of 20, 40, or 80 MHz [6]. The 127 MHz fmax exceeds the highest of these by more than 1.5×, meeting timing on all corners at any realistic operating frequency. Hold slack is positive across all corners.

9 Discussion

The three design-facing verification plan dimensions shaped the implementation in distinct ways. layer produced the module-to-testbench decomposition: each requirement maps to one testable module, and reference models in the sub-module testbenches made white-box sub-claims tractable without exposing RTL internals. format_applicability motivated front-loaded config byte ordering and per-field FSM granularity. side encoded requirements along the ISO transmitter/receiver axis and scopes testbench stimulus to the corresponding drive mode.

The AI-assisted workflow delivered value in two distinct phases, with very different characteristics in each. In the extraction phase, the LLM agent earned its keep by bootstrapping and linking the initial normative statement set. Having a fully populated and linked starting point - even one requiring substantial revision - gave the manual review process a concrete artifact to work from. The time saving from the extraction itself was, however, marginal. The agent’s output had to be reviewed statement by statement, which is

functionally similar to extracting requirements manually in the first place. The primary benefit of the AI-assisted approach in this phase was therefore not efficiency, but rather the increased consistency of an automated pass over the full standard text.

The MCP server interface proved genuinely useful throughout the design, implementation, and verification phases that followed extraction. As implementation decisions were made, requirements were refined - paraphrases sharpened, notes extended, observability classifications updated, and traceability fields populated. All updates were applied using the AI agent through dedicated schema-validated MCP tool calls, each targeting an individual requirement field. The narrow, validated interface made incremental AI-assisted maintenance of the verification plan safe and practical across all three project phases.

Of the nine open requirements, six are LLC requirements deferred pending `can_llc` implementation - a known scope boundary, not a gap in the implemented protocol engine. [REQ-034](#) (P3) is not applicable: this implementation streams frames from LLC to MAC with no shared memory, so the shared memory consistency requirement does not apply. [REQ-035](#) (P2) is an optional operational feature - a node that always signals errors is a fully correct CAN FD participant. [REQ-021](#) (P1) is partially covered: bit-error detection is verified, but sub-claims 2-5 require a frame-aware stimulus source to inject errors at the correct bit positions. A reference model approach closes them (Section 9.2).

The CAN FD stack uses 4,608 logic elements on the Cyclone 10 LP target - 4.0× the 1,146 elements of the existing CC controller (Section 1.1.3). Growth is dominated by the RX frame buffer: at 15 bytes the register-based implementation is unremarkable, but at 70 bytes the write-address decoder and read mux each scale proportionally with buffer depth, becoming the dominant cost in `can_mac_fsm`'s 4,109 LEs. The added protocol features - dual bit rate switching, TDC, three CRC variants, fixed stuffing with SBC, and full fault confinement - account for the remainder. Mapping the 70-byte buffer to a single BRAM instance would eliminate the storage flip-flops and combinatorial decode logic, reducing `can_mac_fsm` to protocol-logic cost (Section 9.2). The worst-case f_{max} of approximately 127 MHz exceeds the highest recommended CAN FD system clock of 80 MHz by more than 1.5× (Section 8.2).

9.1 Objectives Assessment

The three objectives stated in Section 1.4 are assessed against the verification results.

1. **CAN/CAN FD protocol controller in VHDL compliant with ISO 11898-1.** The unified `can_mac_fsm` handles all four in-scope frame formats (CB, CE, FB, FE) in both TX and RX, including dual bit rate switching with TDC ([REQ-024](#), [REQ-025](#)). 28 of 37 requirements are closed. The nine open cases are documented in Section 9.2.
2. **Structured requirements with traceability from ISO 11898-1 to testbench results.** 37 requirements were derived from ISO 11898-1, each linked to its source section, verification method, testbench file, and assertion label. 28 are closed against passing testbenches or code inspection. The full plan is in Section C.
3. **RTL design integrated via Avalon-ST interfaces into Everllence's existing FPGA infrastructure.** The RTL is written in portable VHDL-93 with no vendor primitives. Synthesis confirmed a worst-case f_{max} of 127 MHz, exceeding the highest recommended CAN FD system clock by more than 1.5× at 30% device utilization (Section 8). `can_mac_pcs_fce` exposes a fully functional Avalon-ST TX/RX interface and can be driven directly by the host, following the same integration model as `can_bus_controller`.

9.2 Future Work

1. **can_llc implementation.** The LLC sub-layer is the one unimplemented module. Interface contracts are specified in [REQ-001](#) through [REQ-005](#) and [REQ-036](#). Implementing it adds ISO-specified frame buffering and retransmission, completing the full ISO 11898-1 CAN node.
2. **Hardware integration and bring-up.** The RTL has been verified in simulation only. Bring-up on a physical CAN FD bus would validate timing closure, transceiver compatibility, and bit timing calibration under real bus conditions.
3. **CAN XL support.** CAN XL is out of scope. The layered architecture and unified FSM are well-suited for extension: a third bit rate phase and XL-specific frame format map naturally onto additional PCS rate parameters and new `can_mac_fsm` states.
4. **Frame buffer block RAM migration.** Migrating the 70-byte RX frame buffer to a single M9K block RAM instance would reduce `can_mac_fsm` from 4,109 LEs to protocol-logic-only cost. See [Section 9](#) for the analysis.
5. **Error-type-specific simulation coverage ([REQ-021](#)).** Sub-claims 2-5 require a frame-aware reference model running in parallel with the integrated testbench to track the stuffed bit stream and target injections to the correct frame region.

10 Conclusion

This thesis presented the design, implementation, and verification of a CAN/CAN FD protocol controller in VHDL-93, following the ISO 11898-1 layered reference model. The implemented design covers the MAC, PCS, and FCE sub-layers as independently testable modules, supports all four in-scope frame formats (CB, CE, FB, FE), and implements dual bit rate switching with Transmitter Delay Compensation. 28 of 37 requirements are closed against passing testbenches or code inspection. The nine open cases are documented in [Section 9.2](#). The design uses 4,608 logic elements with a worst-case f_{max} of approximately 127 MHz, exceeding the highest recommended CAN FD system clock of 80 MHz by more than 1.5 $\times$. The ISO 11898-1 layered reference model proved to be a practical partitioning of protocol complexity: each sub-layer was implemented, verified, and debugged independently, a property that motivates Everllence's decision to develop the controller in-house.

11 References

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A Accompanying Digital Materials

The zip file accompanying this document contains the complete source tree developed during this project. The tables below identify the key files by category.

RTL source files

File	Description
src/can_types_p/hdl_src/can_types_p.vhd	Shared types package
src/can_mac/hdl_src/can_mac_fsm.vhd	Unified MAC FSM
src/can_mac/hdl_src/can_mac.vhd	MAC wrapper
src/can_mac_ser/hdl_src/can_mac_ser.vhd	TX serializer
src/can_mac_bs/hdl_src/can_mac_bs.vhd	Bit stuffer/destuffer
src/can_mac_crc/hdl_src/can_mac_crc.vhd	CRC engine
src/can_pcs/hdl_src/can_pcs.vhd	PCS (bit timing, sync, TDC)
src/can_fce/hdl_src/can_fce.vhd	Fault Confinement Entity
src/can_mac_pcs_fce/hdl_src/can_mac_pcs_fce.vhd	Synthesized top-level wrapper

Testbench files

File	Description
src/can_mac_ser/hdl_tb/can_mac_ser_tb.vhd	Serializer
src/can_mac_bs/hdl_tb/can_mac_bs_tb.vhd	Bit stuffer
src/can_mac_crc/hdl_tb/can_mac_crc_tb.vhd	CRC engine
src/can_pcs/hdl_tb/can_pcs_tb.vhd	PCS
src/can_fce/hdl_tb/can_fce_tb.vhd	FCE
src/can_mac_pcs_fce/hdl_tb/can_mac_pcs_fce_tb.vhd	MAC + PCS + FCE integration

Verification plan and tooling

File	Description
verification_plan/verification_plan.toml	37 requirements with traceability metadata
mcp_tools/verification_plan_manager.py	MCP server for verification plan maintenance

B Complete CAN Node Signal Interface

Complete signal-level connectivity of the implemented CAN node. The MAC sub-layer is expanded to show its four constituent entities (can_mac_fsm, can_mac_bs, can_mac_ser, can_mac_crc). PCS and

FCE appear as external module boundaries. The LLC interface block represents the Avalon-ST boundary to the LLC sub-layer, which is not implemented in this project.

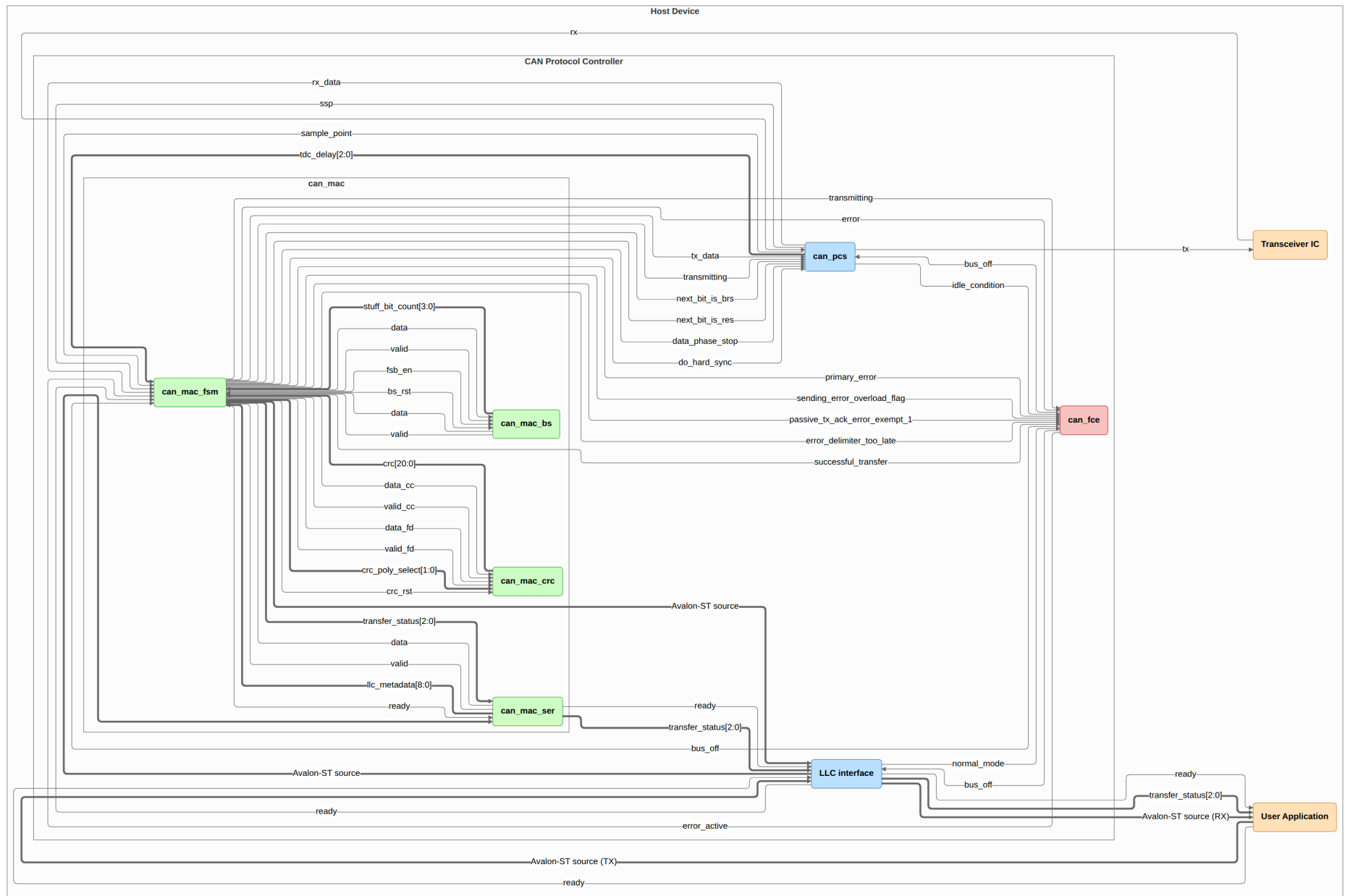


Figure 25: Complete CAN node signal-level connectivity. The MAC sub-layer is expanded to show its four internal entities. The LLC interface block is the Avalon-ST boundary to the pending LLC implementation.

C Verification Plan

Both tables are regenerated automatically from `verification_plan/verification_plan.toml` on each PDF build. The ID field is the join key between them. See Section 2.2 and Section 4 for the meaning of each field. The first table lists each requirement with its ISO source clause, priority, and paraphrase. The second table lists the verification metadata: layer, side, format applicability, observability, method, status, traceability label, file, and coverage criteria.

Table 9: ISO 11898-1 extracted requirements.

ID	ISO ref	Priority	Requirement	Notes
REQ-001	6.4.5.2	P1	LLC notification LLC shall notify the user on every successful DF/RF TX or RX.	Timestamping (§6.5.7) is optional per ISO and is out of scope.
REQ-002	6.4.5.5.2, 6.4.5.5.3	P2	LLC transmission request and abort timing 1) A TX request shall be processed no later than the second SOF after the request when no EFs are present. 2) Frames not yet passed to the MAC are aborted immediately. 3) An abort request shall be processed before the second SOF (no EFs present) or third SOF (with EFs present).	
REQ-003	6.5.2, 6.5.3	P1	Retransmission 1) Frames losing arbitration shall be retransmitted unless aborted by the LLC user. 2) Non-XL nodes shall support unlimited retransmissions.	
REQ-004	6.5.4	P3	Frame acceptance filtering 1) Receivers should use acceptance filtering to decide frame relevance. 2) Filtering should be a single self-contained operation with no state carry-over from previous frames.	
REQ-005	6.5.6	P1	LLC transfer status reporting LLC shall report five transfer status values: Ongoing, Transmitted, Lost_Arbitration, Disturbed, and Aborted, each reflecting the corresponding MAC sub-layer event.	
REQ-006	6.6.4.4	P1	CRC rules 1) CRC_15 for CC frames. CRC_17 for FD frames with payload ≤ 16 B. CRC_21 for FD frames with payload > 16 B. 2) Init vector: all-zero for CRC_15. MSB-one for CRC_17 and CRC_21.	

ID	ISO ref	Priority	Requirement	Notes
REQ-007	6.6.5.2, 6.6.5.3, 6.6.6.2, 6.6.6.3	P1	Error and overload flags 1) Active error and overload flags: 6 consecutive dominant bits each. 2) Passive error flag: 6 consecutive recessive bits (may be overwritten by dominant bits from other nodes). 3) Error and overload delimiters: 8 recessive bits each. 4) After the flag, all nodes monitor the bus and simultaneously begin the remaining 7-bit delimiter on detecting the first recessive bit.	
REQ-008	6.6.7.1, 6.6.7.2, 6.6.7.3, 6.6.7.4, Figure 6, Figure 7	P1	Inter-frame space 1) Intermission shall be exactly 3 recessive bits. 2) No DF or RF may be started during intermission. 3) EFs and OFs shall not be preceded by inter-frame space. 4) Bus idle is recognised at the third recessive intermission bit (error-active nodes/receivers), the eighth recessive suspend bit (error-passive transmitters), or on leaving bus-integration state. 5) An error-passive transmitter shall suspend for 8 bit times after intermission. If another node transmits during the suspend window, the node shall become a receiver.	
REQ-009	6.6.7.5	P1	Bus re-integration 1) Nodes shall start bus re-integration on protocol startup or bus-off recovery. 2) Re-integration completes after 11 consecutive recessive bits at the sample point. 3) For FD-capable nodes, any synchronisation-causing edge shall reset the 11-bit recessive-bit counter.	
REQ-010	6.6.7.3, 6.6.8, Figure 10, Figure 11	P1	Frame initiation 1) A node shall send SOF only when the bus is idle. 2) A dominant third intermission bit causes a node with a pending frame (if error-active or previous receiver) to skip SOF and begin arbitration immediately. 3) A dominant bit detected during bus idle shall be interpreted as SOF.	

ID	ISO ref	Priority	Requirement	Notes
REQ-011	6.6.10.1, 6.6.10.4, Figure 10, Figure 11, Figure 12, Figure 13	P2	Remote frame RF DLC indicates the requested data length of the corresponding DF.	
REQ-012	6.6.10.2, 6.6.11.2, Figure 12, Figure 13, Figure 18, Figure 19	P1	SRR and RRS reserved bits 1) Receivers shall accept recessive and dominant SRR bits without form error. 2) Receivers shall accept recessive and dominant RRS bits without form error.	
REQ-013	6.6.10.5, 6.6.11.5	P1	CRC bit stream coverage 1) CC: SOF, arbitration, control, and data fields (dynamic stuff bits excluded). 2) FD: SOF, arbitration, control, data, dynamic stuff bits, and stuff count field (fixed stuff bits excluded).	
REQ-014	6.6.10.6, 6.6.11.6, 6.6.14, Figure 17	P1	Frame acknowledgement 1) Receivers shall ACK consistent frames and flag inconsistent frames with an EF. 2) Transmitters shall flag unacknowledged frames with an EF. 3) FD frames tolerate up to a 2-bit dominant ACK overlap to compensate for receiver phase differences. 4) ACK shall not depend on the frame identifier.	
REQ-015	6.6.11.3, Figure 20, Figure 21	P2	ESI bit transmission 1) Recessive when the LLC ESI flag is set. 2) Dominant when LLC ESI flag is clear and the node is error-active. 3) Recessive when LLC ESI flag is clear and the node is error-passive.	
REQ-016	6.6.11.5, Table 8	P1	Stuff bit count (SBC) field 1) SBC: 3-bit Gray-coded count of dynamic stuff bits (0-7) plus parity bit, placed before the CRC field. 2) TX shall transmit its dynamic stuff-bit count (up to the first FSB) as the SBC field. 3) RX shall verify the received SBC against its own count.	

ID	ISO ref	Priority	Requirement	Notes
REQ-017	6.6.10.5, 6.6.11.5, Figure 16, Figure 22, Figure 23	P1	CRC delimiter 1) In FD frames a second recessive bit is tolerated before the ACK slot.	
REQ-018	6.6.13.1, 6.6.13.2, 6.6.13.3.1, Table 10	P1	Bit stuffing 1) After 5 consecutive identical bits, insert a complementary dynamic stuff bit. RX discards it. 2) CC: dynamic stuffing spans SOF to FCRC. FD: SOF to end of data field. ACK and EOF are not stuffed. 3) FD CRC field: a fixed stuff bit precedes the SBC field and follows every fourth CRC bit, each the inverse of the preceding bit. No double stuffing. 4) RX discards FSBs and flags a form error if an FSB matches the preceding bit.	
REQ-019	6.6.10.7, 6.6.11.7, 6.6.15.2, Figure 8, Figure 9	P1	End of frame (EOF) 1) A receiver shall validate a frame after the sixth EOF bit. 2) A transmitter requires all seven EOF bits to be error-free.	
REQ-020	6.6.10.2, 6.6.11.2, 6.6.17.4, 6.6.17.5, Figure 10, Figure 11, Figure 12, Figure 13, Figure 18, Figure 19	P1	Bus arbitration Recessive-sent, dominant-monitored: lose arbitration and become a receiver.	

ID	ISO ref	Priority	Requirement	Notes
REQ-021	6.6.21.2	P1	MAC error detection 1) Bit error: sent/monitored mismatch. Exceptions: recessive non-stuff bit during arbitration, recessive bit in ACK slot, dominant bit while sending a passive error flag. 2) Stuff error: sixth consecutive identical bit in a dynamically-stuffed field. 3) Form error: unexpected fixed-form bit value, or FSB not inverse of preceding bit (FB/FE). Exception: dominant last EOF bit or last delimiter bit is not a form error. 4) CRC error: CRC mismatch. FD also requires SBC match and correct parity. 5) ACK error: transmitter detects no dominant bit in the ACK slot.	Sub-claim 1 (bit error) is simulation-verified via recessive injection in test_bus_off. Sub-claims 2-5 require frame-aware error injection not available in the current testbench. All four are covered by code inspection only.
REQ-022	6.6.21.3.1, Figure 28, Figure 29, Figure 30, Figure 31	P1	Error flag initiation and CRC error behaviour 1) On any detected error, start an EF at the next bit and inform the LLC. 2) Data-phase errors: switch to nominal bit rate before starting the EF. 3) CRC error: receiver withholds ACK and sends an EF at the first EOF bit (CC) or 3 bit-times after the CRC delimiter (FD).	
REQ-023	6.6.21.3.2	P1	Overload frame 1) LLC-requested OF: optional, starts at the first intermission bit, at most 2 per inter-frame space. 2) Reactive OF: triggered by a dominant bit at the first two intermission bits, the last EOF bit (receivers only), or the last error/overload delimiter bit. Starts one bit after the trigger.	

ID	ISO ref	Priority	Requirement	Notes
REQ-024	7.3.2, 7.3.3, Table 13, Figure 32	P1	PCS bit timing configuration 1) TQ is a fixed time unit derived from the node clock period. $TQ = m \times t_{q,min}$ where m is a programmable integer prescaler and $t_{q,min}$ = one clock period. 2) A bit time consists of four segments: Sync_Seg (1 TQ), Prop_Seg (compensates for bus propagation and node delays), Phase_Seg1, and Phase_Seg2. The sample point falls at the end of Phase_Seg1. 3) IPT (Information Processing Time): the time a node requires to determine the bit value after sampling. $IPT \leq 2 TQ$. 4) Separate nominal and FD data bit rate configurations, each based on a programmable integer prescaler.	Deployment constraints on the system, not RTL behaviors: prop_seg, SJW, Phase_Seg2, and oscillator tolerance must be programmed to meet the bounds in §7.3.2, §7.3.3, and §7.3.6. Minimum configuration ranges are given in Table 13 (§7.3.3).
REQ-025	7.3.4, 6.6.21.3.1, Figure 34, Figure 35	P1	Transmitter delay compensation 1) TDC applies in the FD data phase when BRS is recessive. Enable and prescaler (1 or 2) are programmable. 2) SSP position = measured transmitter delay + configured offset. 3) TDC active: bit errors at the SP are ignored. SSP errors are flagged at the following SP.	
REQ-026	7.3.5.1, 7.3.5.2, 7.3.5.3, 7.3.5.4, Figure 33	P1	PCS synchronisation 1) Phase error: the time interval between an R-to-D edge and the Sync_Seg boundary of the current bit time. Positive when the edge falls after Sync_Seg (late), negative when before (early). 2) Qualifying edge: R-to-D, previous SP is R. 3) SJW is the maximum per-resynchronization adjustment to Phase_Seg1 or Phase_Seg2. Qualifying edges cause resynchronization. If $ phase\ error \leq SJW$: same effect as hard synchronization. If $phase\ error > SJW$: Phase_Seg1 += SJW. If $phase\ error < -SJW$: Phase_Seg2 -= SJW. 4) One synchronization per bit time, re-enabled after the next R SP. 5) Hard synchronization: IFS edges (except first intermission bit), bus-integration state, and FDF-to-res transition in FD frames. Restarts bit time with Sync_Seg completed.	RTL is stricter than ISO: sync is suppressed unconditionally when transmitting. Safe since the transmitter is the timing source.

ID	ISO ref	Priority	Requirement	Notes
REQ-027	6.6.7.5, 7.4.2.2, 7.4.2.3, 8.1.4.4	P1	PCS bus-off isolation and signalling 1) During bus-off, the PCS shall not drive dominant bits. 2) PCS enters bus-off on a supervisor request and exits on a release request.	
REQ-028	8.1.4.2 rule a), rule b), rule c), rule d), rule e), rule f), rule g), rule h)	P1	Error counter rules 1) REC +1 on any receiver-detected error (except bit errors during error/overload flag). 2) REC +8: first dominant bit after sending an error flag, or bit error while sending an active error/overload flag. 3) REC -1 on successful RX if 1-127. Stays 0 if already 0. Set to 119-127 if above 127. 4) TEC +8 when sending an error flag. Error-passive ACK error is exempt. 5) TEC +8 on bit error while sending an active error or overload flag. 6) TEC -1 on successful TX (floor 0). 7) After 14 consecutive dominant bits post active error/overload flag (8 post passive error flag): both +8. Each additional 8 dominant bits: both +8.	
REQ-029	8.1.3.1, 8.1.3.2, 8.1.4.3, 8.1.4.4, Figure 43	P1	Error confinement state transitions 1) Error-passive: TEC or REC > 127. 2) Error-active: TEC and REC ≤ 127. 3) Bus-off: TEC > 255. 4) Bus-off recovery: 128 idle conditions. TEC and REC reset to zero. 5) FCE notifies LLC on bus-off entry. 6) LLC resets FCE to initial state on request.	Idle condition: 128 × 11 recessive bits monitored
REQ-030	6.6.11.3, Figure 20, Figure 21	P1	BRS bit rate switching 1) Recessive BRS: switch to FD data rate at the BRS sample point. Dominant BRS: keep nominal rate. 2) Rate switches back to nominal at the first CRC delimiter sample point.	

ID	ISO ref	Priority	Requirement	Notes
REQ-031	6.4.3, 6.4.4, Table 5	P1	DLC byte-count mapping DLC 0-8: linear mapping. FD: DLC 9→12, 10→16, 11→20, 12→24, 13→32, 14→48, 15→64 bytes. CC: DLC 9-15 clamped to 8 bytes.	
REQ-032	6.6.16	P1	Byte/Bit transmission order Within the data field byte 0 is sent first. Within each byte bit(7) is sent first.	
REQ-033	8.1.5, Figure 7	P1	Node start-up 1) A node shall complete bus re-integration (11 consecutive recessive bits) before transmitting. 2) A single node receiving no ACK becomes error-passive but never bus-off.	Sub-claim 2 verified via FCE Exception 1 mechanism (passive_tx_ack_error_exempt_1) in can_fce_tb.
REQ-034	6.6.18	P3	MAC data consistency If MAC frames are stored in shared memory, data consistency shall be ensured by buffering the whole frame before transmission starts, or by LLC error-checking during transfer.	Conditional requirement: applies only if shared memory is used. This implementation streams frames directly from LLC to MAC with no shared memory. the LLC is responsible for buffering and retransmission. Not applicable to current architecture.
REQ-035	6.6.21.1, 6.6.21.4	P2	Error signalling enable A configuration parameter shall exist to enable or disable error signalling.	
REQ-036	6.4.4	P3	DLC padding An LLC restricted to fewer bytes shall replace excess bytes with 0xCC padding in both TX and RX directions.	Only applicable if the implementation exposes a configurable maximum-data-byte restriction. If not implemented, waive.

ID	ISO ref	Priority	Requirement	Notes
REQ-037	5.2, Figure 2	P1	Frame field sequence 1) CBFF DF: SOF (D) → Arbitration (ID[28:18], RTR (D)) → Control (IDE (D), r0 (D), DLC[3:0]) → Data → CRC (CRC_15, delimiter (R)) → ACK (slot, delimiter (R)) → EOF (7×R). 2) CBFF RF: as CBFF DF but RTR (R) and no data field. 3) CEFF DF: SOF (D) → Arbitration (ID[28:18], SRR (R), IDE (R), ID[17:0], RTR (D)) → Control (r1 (D), r0 (D), DLC[3:0]) → Data → CRC (CRC_15, delimiter (R)) → ACK (slot, delimiter (R)) → EOF (7×R). 4) CEFF RF: as CEFF DF but RTR (R) and no data field. 5) FBFF DF: SOF (D) → Arbitration (ID[28:18], RRS (D)) → Control (IDE (D), FDF (R), res (D), BRS, ESI, DLC[3:0]) → Data (FD rate) → CRC (SBC[3:0], CRC_17/21, delimiter (R)) → ACK (slot, delimiter (R)) → EOF (7×R). 6) FEFF DF: SOF (D) → Arbitration (ID[28:18], SRR (R), IDE (R), ID[17:0], RRS (D)) → Control (FDF (R), res (D), BRS, ESI, DLC[3:0]) → Data (FD rate) → CRC (SBC[3:0], CRC_17/21, delimiter (R)) → ACK (slot, delimiter (R)) → EOF (7×R).	Notation: (D) = dominant, (R) = recessive. Unannotated bits (ID, DLC, BRS, ESI, CRC value, SBC, ACK slot) are data-dependent or conditional.

Table 10: ISO 11898-1 verification plan.

ID	Layer	Side	Format	Observability	Method	Status	Label	File	Coverage criteria
REQ-001	LLC	both	CB, CE, FB, FE	black_box	simulation	not_started			
REQ-002	LLC	transmitter	CB, CE, FB, FE	white_box	code_inspection	not_started			
REQ-003	LLC	transmitter	CB, CE, FB, FE	white_box	code_inspection	not_started			
REQ-004	LLC	receiver	CB, CE, FB, FE	white_box	code_inspection	not_started			
REQ-005	LLC	transmitter	CB, CE, FB, FE	black_box	simulation	not_started			
REQ-006	MAC	both	CB, CE, FB, FE	white_box	simulation, coverage	complete	crc_15, crc_17, crc_21	can_mac_crc_tb	Three bins: (1) CB or CE frame (CRC_15), (2) FB or FE frame with data ≤ 16 B (DLC ≤ 10 , CRC_17), (3) FB or FE frame with data > 16 B (DLC ≥ 11 , CRC_21) - must each be hit.
REQ-007	system	both	CB, CE, FB, FE	white_box	code_inspection, wave-form_inspection	complete	p_fsm:~900, test_bus_off	can_mac_pcs_fce_tb	

ID	Layer	Side	Format	Observability	Method	Status	Label	File	Coverage criteria
REQ-008	MAC	both	CB, CE, FB, FE	white_box	code_inspection, wave-form_inspection	complete	p_fsm:~486, test_normal, test_bus_off	can_mac_pcs_fce_tb	
REQ-009	system	both	CB, CE, FB, FE	white_box	code_inspection, wave-form_inspection	complete	p_fsm:~433, test_bus_off	can_mac_pcs_fce_tb	
REQ-010	MAC	both	CB, CE, FB, FE	white_box	code_inspection, wave-form_inspection	complete	p_fsm:~451, test_normal	can_mac_pcs_fce_tb	
REQ-011	MAC	both	CB, CE	black_box	simulation	complete	FTYP Coverage	can_mac_pcs_fce_tb	
REQ-012	MAC	both	CE, FB, FE	white_box	code_inspection, wave-form_inspection	complete	p_fsm:~546, p_fsm:~548, test_normal	can_mac_pcs_fce_tb	
REQ-013	MAC	both	CB, CE, FB, FE	white_box	code_inspection	complete	p_fsm:~299, p_fsm:~347, p_fsm:~983	can_mac_fsm	
REQ-014	system	both	CB, CE, FB, FE	white_box	simulation, wave-form_inspection	complete	test_normal	can_mac_pcs_fce_tb	
REQ-015	MAC	transmitter	FB, FE	white_box	simulation, wave-form_inspection	complete	test_normal	can_mac_pcs_fce_tb	
REQ-016	MAC	both	FB, FE	black_box	simulation	complete	p_sbc_checker	can_mac_bs_tb	
REQ-017	MAC	transmitter	CB, CE, FB, FE	white_box	code_inspection, wave-form_inspection	complete	p_fsm:~768, p_fsm:~796, test_normal	can_mac_pcs_fce_tb	
REQ-018	MAC	both	CB, CE, FB, FE	white_box	simulation, coverage	complete	Input Coverage, Output Coverage	can_mac_bs_tb	All bins in cov_input and cov_output must be hit.
REQ-019	MAC	both	CB, CE, FB, FE	white_box	simulation	complete	test_normal	can_mac_pcs_fce_tb	

ID	Layer	Side	Format	Observability	Method	Status	Label	File	Coverage criteria
REQ-020	system	transmitter	CB, CE, FB, FE	black_box	simulation	complete	test_lost_arb	can_mac_pcs_fce_tb	
REQ-021	MAC	both	CB, CE, FB, FE	white_box	code_inspection, wave-form_inspection	in_progress	p_fsm:~283, p_fsm:~336, p_fsm:~343, p_fsm:~415, test_bus_off	can_mac_pcs_fce_tb	
REQ-022	MAC	both	CB, CE, FB, FE	white_box	code_inspection, wave-form_inspection	complete	p_fsm:~283, p_fsm:~294, p_fsm:~357, test_bus_off	can_mac_pcs_fce_tb	
REQ-023	MAC	both	CB, CE, FB, FE	white_box	code_inspection	complete	p_fsm:~370, p_fsm:~384, p_fsm:~944	can_mac_fsm	
REQ-024	PCS	both	CB, CE, FB, FE	white_box	simulation	complete	test_normal	can_pcs_tb	
REQ-025	PCS	transmitter	FB, FE	black_box	simulation	complete	p_check_tdc_delay	can_pcs_tb	
REQ-026	PCS	both	CB, CE, FB, FE	white_box	code_inspection, wave-form_inspection	complete	p_can_pcs:~177, p_can_pcs:~228, test_normal	can_pcs_tb	
REQ-027	PCS	both	CB, CE, FB, FE	white_box	simulation, code_inspection	complete	p_can_pcs:~280, p_can_pcs:~379, test_bus_off	can_mac_pcs_fce_tb	
REQ-028	FCE	both	CB, CE, FB, FE	black_box	simulation	complete	test_rule_a, test_rule_b, test_rule_c, test_rule_c_exception, test_rule_d, test_rule_e, test_rule_f_tx, test_rule_f_rx, test_rule_g, test_rule_h	can_fce_tb	Rule h: the '>127 → 119-127' clamp path must be explicitly confirmed in simulation.
REQ-029	FCE	both	CB, CE, FB, FE	black_box	simulation	complete	test_reset, test_bus_off, test_bus_off_recovery	can_fce_tb	

ID	Layer	Side	Format	Observability	Method	Status	Label	File	Coverage criteria
REQ-030	MAC	both	FB, FE	white_box	code_inspection, wave-form_inspection	complete	p_fsm:~636, p_fsm:~755, test_normal	can_mac_pcs_fce_tb	
REQ-031	MAC	both	CB, CE, FB, FE	black_box	simulation, coverage	complete	dlc_cov; test_normal	can_mac_ser_tb.vhd; can_mac_pcs_fce_tb	Four non-linear FD DLC values (9, 12, 14, 15 → 12, 24, 48, 64 bytes) plus DLC=8 boundary must each produce a correctly sized frame.
REQ-032	MAC	both	CB, CE, FB, FE	black_box	simulation	complete	test_normal	can_mac_pcs_fce_tb	
REQ-033	system	both	CB, CE, FB, FE	black_box	simulation	complete	test_rule_c_exception; normal_test, test_bus_off (waveform)	can_mac_pcs_fce_tb	Sub-claim 1: confirm no TX before 11 consecutive recessive bits are detected at the SP. Sub-claim 2: single-node scenario with no ACK injection; TEC must reach error-passive but not exceed 255.

ID	Layer	Side	Format	Observability	Method	Status	Label	File	Coverage criteria
REQ-034	MAC	transmitter	CB, CE, FB, FE	white_box	code_inspection	not_started			
REQ-035	MAC	both	CB, CE, FB, FE	white_box	code_inspection	not_started			
REQ-036	LLC	both	CB, CE, FB, FE	black_box	simulation	not_started			
REQ-037	MAC	both	CB, CE, FB, FE	white_box	waveform_inspection	complete	test_normal	can_mac_pcs_fce_tb	